



Behavioral biometrics & continuous user authentication on mobile devices: A survey

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ABSTRACT

This paper offers an up-to-date, comprehensive, extensive and targeted survey on Behavioral Biometrics and Continuous Authentication technologies for mobile devices. Our aim is to help interested researchers to effectively grasp the background in this field and to avoid pitfalls in their work. In our survey, we first present a classification of behavioral biometrics technologies and continuous authentication for mobile devices and an analysis for behavioral biometrics collection methodologies and feature extraction techniques. Then, we provide a state-of-the-art literature review focusing on the machine learning models performance in seven types of behavioral biometrics for continuous authentication. Further, we conduct another review that showed the vulnerability of machine learning models against well-designed adversarial attack vectors and we highlight relevant countermeasures. Finally, our discussions extend to lessons learned, current challenges and future trends.

1. Introduction

For several decades, user authentication was based on the “something the user knows” paradigm referred as knowledge-based authentication method [139]. This method has been the most popular for authenticating an individual, but research has shown that PINs and passwords do not provide adequate protection [13,140]. Especially when the method is used on mobile devices it is known to suffer from low user-friendliness and insufficient security [102]. Also, mobile devices are vulnerable to smudge attacks and PINs and secret touch patterns can be revealed [65]. Hence, the theft of a device may give rise to the risk of allowing full access to critical applications and personal data. In addition, Stylios et al. [58], showed that a high percentage (24%) of smartphone users ignore privacy and security risks and store, on their mobile devices, large volumes of private information including: PINs, credit card numbers, etc. Also, 19.1% of users ignore protection practices for their PINs and passwords. These vulnerabilities stress the need for the development and implementation of novel authentication methods. These methods are based on “something that the user is” paradigm, that is something that characterizes the user and constitutes a unique morphological biometric feature (e.g., fingerprint, iris, etc.). Under this paradigm, authentication is performed by comparing a

previously captured biometric template to a biometric feature of the person (e.g. a fingerprint) [3,12,44,97,152]. The previously captured biometric template must derive from the same person and be of the same type [141].

Whether based on PINs and passwords or on biometrics, these methods use the entry-point authentication model which authenticates the user only at the beginning of the session [33]. The session authentication model has been heavily criticized since it becomes vulnerable to attacks that occur after the initial authentication [1,5,33]. For these reasons, a new method of user authentication, also based on “something that the user is” paradigm, has been proposed. This method uses Behavioral Biometrics (BB) and Continuous Authentication (CA) [1,97]. As mobile devices become more advanced technologically, it is abundantly clear that the incorporated sensors can be used to efficiently capture the behavior of most users, thus enabling behavioral biometric user authentication [54,55,84,86,114]. Mobile devices can enroll BB templates from their sensors [86,109,141]. BB can include walking gait, touch gestures, keystroke dynamics, hand waving, user profile and power consumption. BB have the advantage of using something that is just as unique as the individual and provide continuous authentication [141]. CA technology constitutes an additional security measure alongside the initial login process by monitoring user behavior and

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continuously re-authenticating user identity throughout a session [85, 94, 97, 116, 117]. The idea of CA occurred in the early 2000s. The interest in this technology has increased since then, both within the academia and the industry [116]. Increased interest in BB/CA technologies is fueled by the expected reduction in the cost of sensors, the improvement of systems and the sociopolitical pressure for better security controls [6]. Finally, research like that presented in [5, 7, 8] showed that users are eager to adopt biometric authentication methods to protect their privacy.

1.1. Motivation

The scope of our survey is wide, ranging from reviewing BB and CA technologies to data collection methodologies and relevant machine learning models. The main goal of our survey is to offer a sufficient background on BB/CA technology for mobile devices, of interest to both researchers and practitioners. Our purpose is to present all the significant elements for enabling researchers to conduct their research. The first goal is to present a classification of all seven categories of BB and CA on mobile devices and an analysis of BB collection methodologies, feature extraction and the existing publicly available datasets. The second goal is to present a literature review on machine learning models performance in seven types of BB for CA. Further, we conduct another review that showed the vulnerability of machine learning models against well-designed adversarial attack vectors and we highlight relevant countermeasures. Finally, our discussions extend to lessons learned, current challenges and future trends.

1.2. Relevant surveys

Recently published surveys, such as Crawford [87] and Yampolskiy & Govindaraju [89], refer to a single behavioral biometric. More specifically Crawford refers to keystroke dynamics and Yampolskiy & Govindaraju to behavioral profiling. Similarly, Ashbourn [91] refers to keystroke dynamics. Ferrag et al. [154] present electrocardiographic signal (ECG), keystroke and touch dynamics while Rui et al. [155] also consider voice recognition. Jain et al. [92] name two modalities used in CA, that is walking gait and keystroke dynamics, while Rogowski et al. [93] also consider the modality of touch gestures. Meng et al. [9] present four CA behavioral biometrics and also refer to morphological biometrics or behavioral biometrics not used for CA. Mahfouz et al. [11] present touch and keystroke dynamics, behavioral profiling and gait recognition. Finally, Abdulaziz and Kalita [94] provide a more extensive list of CA behavioral biometrics, including walking gait, touch gestures, keystroke dynamics, hand waving, and behavioral profiling.

The existing aforementioned surveys are quite old; thus, they contain very few references to recent literature. In addition, there is no reference to attack points on behavioral biometrics and countermeasures except for the work of Meng et al. [9]. Therefore, there is a striking lack of an up-to-date, thorough, exhaustive, and focused survey on CA and BB that would include the previously mentioned issues. Our survey attacks exactly these topics focusing on BB and CA approaches amenable to mobile devices. Our goal is to cover all BB for CA categories, attack vectors and defense techniques.

1.3. Contribution

The contribution of this paper can be summarized as follows:

- We present a classification on seven categories of behavioral traits and an analysis of behavioral biometrics collection methodologies and feature extraction and the existing publicly available datasets.
- We present a recent state-of-the-art literature review on BB/CA technology and the performance of machine learning models.
- We present a literature review on possible attack vectors on BB/CA technology and highlight promising countermeasures.

- We identify the challenges and future trends.

2. Classification and analysis

In this section, we present a classification of BB and CA on mobile devices into seven categories. The first category is walking gait, the second category is touch gestures, the third category is keystroke dynamics, the fourth category is behavioral profiling, the fifth category is hand waving, the sixth category is power consumption and the seventh category is fusion. We also offer an analysis of behavioral biometrics collection methodologies and feature extraction and the existing publicly available datasets. All behavioral traits can be categorized in these seven categories:

2.1. Walking gait

This recognition technique is comparatively new and is based on the measurement and analysis of an individual's manner of walking or running by using the acceleration signals produced by his/her mobile device. The characteristic of gait is amenable to smartphones due to their incorporated sensors, namely accelerometer, gyroscope and magnetometer. The main advantage of this technique is that it can be applied to CA of users without requiring their intervention. However, factors such as the changing orientation of the device during walking [132], uneven ground, possible injuries, footwear, fatigue, personal peculiarities, etc. can affect its accuracy [86].

- **Data Collection:** The accelerometer is used to collect data when subjects are walking in normal, slow, and fast speeds. For estimating the smartphone's orientation in the participants' pocket, gyroscope data is used. Finally, the combination of accelerometer, gyroscope and magnetometer data allows for calculating the motion patterns of the human body [86].
- **Feature extraction:** The features used in the literature of walking gait analysis are speed, orientation, and human body motion patterns. The sensors in use are the accelerometer, the magnetometer and the gyroscope which respectively measure along the local X , Y , Z axes of the device the acceleration in m/s^2 , the ambient geomagnetic field in μTesla and the rate of rotation in rad/s [86, 94, 103].

2.2. Touch gestures

The touch gestures are shapes drawn by hand on the mobile devices touch screen that comprise of a single, or multiple strokes. Each stroke is a series of successive numerical coordinates. Features such as touch direction and the touch duration, the velocity and acceleration of movement are analyzed and measured solely or in combination with each other.

- **Data Collection:** The smartphone touch screen sensor is used for collecting touch data. Actions of input associated with parameters such as speed, velocity, size, length, direction or pressure, are converted into a gesture output template. These parameters are different between users and indicate the behavior of each, thus consisting the basis of the touch gesture authentication systems [86].
- **Feature extraction:** The collected data from users are extracted and analyzed to identify each user by his distinctive set of features. When the finger touches the mobile device screen whether it strokes or swipes it produces a series of touch data as shown in the following formula [84]:

$$S_i = (x_i, y_i, t_i, p_i, A_i, O_i^f, O_i^{ph}), i = \{1, 2, \dots, N\}, (1)$$

where x_i, y_i are the location points, t_i are the time stamps, p_i is the finger pressure on screen, A_i is the finger blocked area, O_i^f is the finger orientation and O_i^{ph} is the device orientation (portrait or landscape) and N represents the total number of swipes.

2.3. Keystroke dynamics

The procedure of recording the typing keyboard inputs of an individual on a mobile device and the effort to identify him via an analysis based on his tapping habits is called keystroke dynamics [11,16].

- **Data collection:** Some researchers on keystroke dynamics collect data from predefined texts, for example during the typing of a text message, or during the log-in session when entering passwords [40, 41, 43, 110, 111]. Others conduct their research by collecting data not restricted on predefined sentences or passwords [42, 44], [44]]. In both cases the results are of high accuracy.
- **Extracted features:** Duration, latency, pressure and location are the features used in the literature of keystroke analysis as explained following [105, 110]:
 - *Duration* is also called hold time and corresponds to the period of time elapsed between the action of pressing and the action of releasing a key.
 - *Latency* is also called inter-key time and is expressed by the time period elapsed between the action of releasing a pressed key until pressing the next key.
 - *Pressure* is the pressure on a key.
 - *Location* are the finger location points (x_i , y_i) or the finger area on screen.

2.4. Behavioral profile

The usage data of a mobile device can be employed for the behavioral authentication of individuals on the basis that they usually follow a specific pattern when using their phones to interact with applications and digital services [2]. The behavioral profile of a user can be built either based on his interaction with a network or with a host. In the first case the behavior of users is monitored regarding their patterns of connecting to Wi-Fi networks, service providers, etc., while in the second case the monitoring refers to the usage manner of applications at different locations and time [94].

- **Data Collection:** Device usage data can be used in several combinations for the profiling of users. For example, Anjomshoa et al. [115] have used a self-developed behavioristic mobile application, namely TrackMaison to collect usage data resulting from the interaction with five social network services such as location and session duration and usage frequency. Neil et al. [50] have used Wi-Fi, Bluetooth and application usage data. Fridman et al. [118] used the device location, applications used, the text entered, and the websites visited. Ashibani et al. [156] used the interaction patterns of users while connecting to their home network, via a hub, to access and control IoT devices from their smartphones. Their model authenticated users based on the Android applications' access logs. In another study of Ashibani and Qusay [169] they authenticated users based on the events of mobile device application access in a smart home environment. Lee et al. [157] used SmarterYou, a system that combined three devices, a smartphone, a smartwatch, and an authentication server in the cloud. The smartwatch was used to monitor the raw data from the accelerometer and the gyroscope sensors and send the information to the smartphone via Bluetooth. The server in the cloud was used for computations and protection of data. active authentication based on multiple biometric and behavior-based profiling systems. Acien et al. [158], used the combination of touch gestures, keystrokes, applications usage data, WiFi, and GPS location while users interacted naturally with their smartphones. They examined two scenarios depending on the number of sessions. The first scenario included one session (One-Time Authentication), while the second scenario included multiple sessions (Active Authentication).

- **Feature extraction:** In the work of Anjomshoa et al. [115] the Track Maison application collected spatiotemporal information regarding the usage of social network services. In addition, when TrackMaison was recording a session, it utilized the incorporated GPS of the device and recorded the location coordinates where the session initiated. Periodically it updated users' location. In the dataset used by Neil et al. [50] all data types included the date and time of data capture and the identification number of users. For the application usage data, the available information additionally included the name of the accessed application. For the Bluetooth usage data, the available information further included the name of the sighted Bluetooth device, the Received Signal Strength (RSSI - dBm) and the MAC address. For the Wi-Fi usage data the available information additionally included the SSID (network name), the Received Signal Strength (RSSI - dBm) and the Access Point's MAC address. Regarding the four modalities studied by Fridman et al. [118], they counted the action's instances related to each modality of TEXT, APP, WEB and Location. Ashibani et al. [156] concentrated on the usage patterns of Android applications which consisted of the length and time of activity when users were accessing the network. Ashibani and Qusay [169] used the smart home hubs to collect the application access history including user id, timestamp, and name of the package. Lee et al. [157] used the accelerometer and the gyroscope sensors of the smartphone and the smartwatch to record the walking patterns and the manner of holding the smartphone. Regarding the application usage data, WiFi, and GPS location, Acien et al. [158] used timestamps and the frequency of the events, while for touch gestures they used the data of x , y , and z coordinates of the gyroscope and the acceleration vector from the accelerometer in each time stamp. For keystroke dynamics, they used hold time, press-latency, and press-release latency.

2.5. Hand waving

The waving pattern of the wrist of an individual, while interacting with his mobile phone or just holding it, for identifying users has recently gained attention. This method does not require any additional action by the user besides holding the device. Several features can be used such as wrist twisting, speed, waving range and frequency. Individuals can be distinguished since the waving of their hand is different [97].

- **Data Collection:** Sitová et al. [38] captured Hand Movement, Orientation, and Grasp (HMOG) by recording the readings from the accelerometer, gyroscope and magnetometer sensors. They captured the device orientation and subtle movement deriving from the manner an individual taps on the smartphone after grasping and while holding it. Yang et al. [100] selected the three-axis accelerometer sensor to measure the phone's acceleration during the waving action. Buriro et al. [98] collected the movement patterns of hand (s) resulting from the accelerometer, gyroscope, magnetometer, orientation and gravity sensors to profile users.
- **Feature extraction:** The features of grasp resistance and grasp stability can be extracted when users tap on the screen [38]. The resistance of a hand holding the device during a tap is measured by the grasp resistance features while the amount of time elapsed for the influence of movement caused by the tap to disappear is measured by the grasp stability features. The grasp resistance and grasp stability features can be extracted from the accelerometer, the gyroscope, and the magnetometer sensors (magnitude, x , y , and z axes). Also, some common features in hand waving extracted from each sensor are median, mean, standard deviation, mean absolute deviation, unbiased standard error of the mean, unbiased skewness and kurtosis [98].

2.6. Power consumption

The fact that patterns of user behavior are highly correlated to the patterns of power consumption constitute the basis of this approach [129]. To the best of our knowledge power consumption is not used as a single identifier on biometric systems but in combination with other biometrics [86].

- **Data Collection:** Murmura et al. [86] measured the power consumption resulting from the activities performed by the user by using the incorporated sensors of the battery driver of mobile devices namely the voltage and current sensors. In another work of Murmura et al. [130], they collected information regarding the power usage from the power suspend driver. They recorded the power usage of the CPU, display, graphics, audio, microphone, Wi-Fi, and GPS.
- **Feature extraction:** The voltage is reported by the sensors to the kernel of the operating system in units micro-volts (uV) and the current in micro-Ampere (uA). The battery charge is reflected on the voltage readings while the quantity drawn by the Operating System is reflected on the current readings and depends on the kind of activities performed [86]. The power-management module of the operating system reported the use of each subsystem's time shares. Based on these measurements, they calculated the power consumption of each application [130].

2.7. Fusion

In this section we will analyze the fusion achieved by a single sensor using multiple instances of a biometric and the fusion achieved by multiple sensors. The focus will be made on multimodal behavioral biometrics fusion. The purpose of multimodal systems is to access multiple sources of information from different modalities and improve the authentication accuracy [23]. Several researchers have used multi-modal behavioral biometrics traits, such as touch gestures, hand waving, behavioral profile, keystroke dynamics, walking gait, etc., to authenticate users [1,24,25]. According to Sanderson and Paliwal [14] fusion levels are divided into two categories:

- Pre-classification or fusion before matching:** Pre-classification refers to fusion at the sensor level (or raw data) and the feature level.
 - Sensor-level fusion is the unification of raw data obtained using multiple sensors or multiple instances of a biometric using a single sensor.
 - In feature-level fusion, the feature sets originating from multiple biometric algorithms are unified into a single feature set.
- Post-classification or fusion after matching:** Post-classification refers to fusion at the match score, rank and decision levels.

In match score-level fusion the combination of the match scores output resulting from multiple biometric matchers generates a new match score, which is thereafter used by the authentication system to achieve a decision. When a biometric system operates, the ranking of the enrolled identities constitutes the system's output. This output expresses the sorting of all possible sets of matching identities in decreasing order of confidence. In decision level fusion the majority voting approach is mostly used where the input of a biometric sample is attributed to that identity on which the majority of the matchers agree [20]. There is a large corpus of research works referring to fusion. More information can be found in the following works: [17–22,90,108].

2.8. Publicly available datasets

There are some publicly available datasets like the one in the work of Mahbub et al. [166] that can be used for research purposes. This dataset was designed for biometric identification and it contains smartphone

sensor signals collected from 48 volunteers over a period of 2 months. The data collected include the front-facing camera, touchscreen, gyroscope, accelerometer, magnetometer, light sensor, GPS, Bluetooth, WiFi, proximity sensor, temperature sensor, and pressure sensor. The authors also stored the timing of screen lock and unlock events, start and end timestamps of calls, currently running foreground applications, etc. Another publicly available dataset is the itekube – 7 touch gestures dataset which was created from Debard et al [106] and it contains 6591 touch gestures of 7 different interaction classes.

Other publicly available datasets can be found in the works of Reyes-Ortiz et al. [164] and Casado et al. [165] which, although were not initially intended for biometric identification they could be used for that purpose. The dataset of Reyes-Ortiz et al. [164] was originally proposed for the recognition of human activity and it contains raw data from the inertial sensors (accelerometer and gyroscope) of a smartphone when carried by 30 different users. The dataset provided by Casado et al. [165] contains raw data from the accelerometer, gyroscope, and magnetometer of a mobile device when carried by 77 different people.

2.9. Evaluation

In this section we present the performance metrics and the classifiers that are used in section 3 that follows.

2.9.1. Performance metrics

The basic metrics that are applied for evaluating an authentication system depend on the error rates. In the following, we discuss some basic metrics used in calculating authentication errors [12,127]:

- The *Confusion matrix* is used to describe the performance of a classifier. In Table 1 we present an example of a confusion matrix where there are two possible predicted classes: "Genuine" and "Impostor".

Where:

- *TA (True Acceptance)* is the number of patterns belonging to the genuine user and ranked correctly as "Genuine".
- *TR (True Reject)* is the number of patterns not belonging to the genuine user and ranked correctly as "Impostor".
- *FA (False Acceptance)* is the number of patterns not belonging to the genuine user and ranked mistakenly as "Genuine".
- *FR (False Reject)* is the number of patterns belonging to the genuine user and ranked mistakenly as "Impostor".

Based on the above, the True Acceptance Rate (TAR), False Acceptance Rate (FAR), False Reject Rate (FRR), Accuracy and Equal Error Rate (EER) are estimated as follows [127]:

- *True Acceptance Rate (TAR)* is the conditional probability of a pattern to be classified in the class "Genuine" given that it belongs to it. TAR is given by the formula:

$$TAR = \frac{TA}{TA + FR} \quad (7)$$

- *False Acceptance Rate (FAR)* is the conditional probability a pattern to be classified in the class "Genuine" given that it does not belong to it. FAR is given by the formula:

Table 1
Confusion matrix.

		Actual Class	
		Genuine	Impostor
Predicted Class	Genuine	TA	FA
	Impostor	FR	TR

$$FAR = \frac{FA}{FA + TR}, \quad (8)$$

- **False Reject Rate (FRR)** is the conditional probability a pattern not to be classified in the class “Genuine” given that it belongs to it. FRR is given by the formula:

$$FRR = \frac{FR}{FR + TA}, \quad (9)$$

- **Accuracy** is defined as the probability of a correct classification of a pattern. Accuracy is given by the formula:

$$Accuracy = \frac{TA + TR}{TA + TR + FA + FR}, \quad (10)$$

- **Equal Error Rate (EER)** is the error rate that is achieved by tuning the detection threshold of the system such that FAR and FRR are equal [170]. (11)

In addition, the following metrics are often used:

- **Decision threshold:** the claimed identity is considered genuine, thus accepted, when its matching score exceeds a predefined threshold $\alpha \in (0, 1)$. In case the matching score of the claimed identity does not exceed the predefined threshold, it is perceived as an impostor, thus rejected. The formula for calculating the decision rule is the following [11]:

$$P(ui) = \begin{cases} \text{Genuine}, & P(ui) \geq \alpha \\ \text{Impostor}, & P(ui) < \alpha \end{cases}, \quad (12)$$

where α expresses a predefined threshold and $P(ui)$ expresses the score for a user ui authentication.

- **Receiver Operating Characteristic (ROC) Curve** demonstrates the performance based on TAR and FAR. The ideal point is represented by the top left corner where TAR equals one and FAR equals zero [11, 94]. The **Area Under Curve (AUC)** is used as an alternative to accuracy and the values range between 0 and 1 [11].
- **Mean Absolute Error (MAE):** the average of all absolute errors, described in the following formula [4]:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (13)$$

Where: n = the number of errors, Σ = summation, $|y_i - \hat{y}_i|$ = the absolute errors.

- **Mean Squared Error (MSE):** measures the average squared error of model's predictions. where y_i is the expected output and $\hat{y}_i$ is the prediction of model [4]:

$$MSE = \frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2, \quad (14)$$

- **Root mean squared error (RMSE):** is the square root of the average of squared differences between prediction and observation [4]:

$$RMSE = \sqrt{\frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2}, \quad (15)$$

- **Half Total Error Rate (HTER):** is equal to the aggregation of FAR with FRR divided by 2. Decreased HTER, signifies better system performance. The HTER is calculated with the following formula [99]:

$$HTER = \frac{FAR + FRR}{2}, \quad (16)$$

- **Manhattan distance (Rectilinear distance L1):** is the sum of the lengths of the projections of the line segment between the points onto the axes of coordinates [107]:

$$d(a, b) = \sum_{i=1}^n |a_i - b_i|, \quad (17)$$

- **Euclidean distance L2:** is the length of the line segment connecting points a and b ($\overline{ab}$) [107]:

$$d(a, b) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}, \quad (18)$$

- **Dynamic Time Warping (DTW) distance metric:** We can shortly describe it as an algorithm that can align temporal sequences and measure their similarities. More information can be found in the work of Zhao et al. [171].

2.9.2. Classifiers and machine learning algorithms

The purpose of this paper is not to analyze each classifier; we solely report them with a citation, in case someone is seeking for more information to easily find it. The algorithms that we have encountered are: Random Forests [61], Support Vector Machines (SVM) [62], Bayesian Networks [63], J48 tree [63], Multi-layer Perceptron (MLP) [63], Naïve Bayes [63], Instance-Based Learning Algorithms [48], Dynamic Time Wrapping (DTW) [66], k-Nearest Neighbors (kNN) [135], Long Short-Term Memory (LSTM) [88], Logistic Regression used as a classifier [136], Cycle detection [132], Manhattan method [107], isolation Forest [26], Hidden Markov Model (HMM) [27], Convolutional Neural Networks (CNNs) [106], Feed Forward Neural Network (FFNN) [28], Generalized Regression Neural Networks (GRNNs) [40], Least Squares Anomaly Detection (LSAD) [42], Radial Basis Function networks (RBF) [46], Modified Edit-Distance algorithm (M-ED) [119], Ordinary Least Squares regression (OLS) [130] and Strangeness-based Outlier Detection algorithm (strOUD) [86].

2.10. Discussion on classification and analysis

One major challenge for BB research is the small number of real-world datasets and the availability of a public behavioral biometrics database for research purposes [10]. There are some available datasets as in the work of Mahbub et al. [166] and Debar et al [106] that can be used for research purposes. Other publicly available datasets can be found in the works of Reyes-Ortiz et al. [164] and Casado et al. [165] which, although were not initially intended for biometric identification, could all be used for that purpose. Also, many users avoid participating in time consuming, painstaking procedures for the collection of biometric features. This results in not fulfilling the data collection procedure. For this reason, the testing of a methodology for extracting biometric features, in a user-friendly way, constitutes an open problem. Also, BBCA technology is of limited extent due to some fundamental

deficiencies, as, the Biometrics Privacy Concerns (BPC) of users [1]. There are two approaches to data collection and data processing. In the first approach, which is device-centric, data is collected and processed in the device itself, while in the second approach, data is sent to a central online server. Users may be concerned about their sensitive information and thus, we believe that research should focus on ways to preserve user privacy. This can be achieved by processing behavioral measurements on the user's mobile and that sensitive information is not sent to the online service. Also, in the work of Shila et al [102], it was shown that the device-centric implementation of authentication outperformed the cloud-based authentication in terms of classification accuracy and detection latency. In addition, particular attention should be given to the anonymization of behavioral biometrics data when these are collected for research purposes. If reliable and effective anonymization is not achievable, then specific permission or consent should be obtained, in accordance with applicable law.

The most important metrics are FAR, TAR, FRR, EER, and Accuracy. In systems that we are interested in security, we want a small FAR and we can sacrifice the FRR to achieve low FAR. In systems that we are interested in friendliness, we want the opposite. TAR is actually the reverse of FAR. EER and Accuracy are more generic metrics that show the overall performance of the algorithm. EER is the error rate that is achieved by tuning the system's detection threshold thus much that FAR and FRR are equal. Accuracy is the percentage of samples that are classified accurately [170]. There are more specialized indicators such as ROC, MAE, HTER, RMSE, etc. The ROC curve demonstrates the dependency between the FAR, FRR, and detection threshold of the system. It is used to show how threshold values affect the overall accuracy of the algorithm. The Half Target Error Rate (HTER) is the average between the FAR and FRR at a random threshold. MAE measures the average magnitude of the errors in a set of predictions, without considering their direction. It is used as a combined metric of FAR and FRR. RMSE is the square root of the average of squared differences between prediction and observation [4]. It is used to estimate how the model fits the data and it is an alternative for MSE which is used for the same purpose. Research works should include at least the FAR, TAR, FRR, EER, and Accuracy when evaluating the performance of their approach. In this way, a comparison between the different approaches will be feasible. Finally, when an authentication approach is evaluated, researchers should also weight and report the amount of data from each user that is necessary for each approach to be effective, the computational and communication cost, and the usage of battery, memory, and CPU. We believe that future research should also measure those aspects to enable the comparison.

3. A review on machine learning authentication models for BBCA

Behavioral biometrics authentication techniques are built based on an individual's behavioral characteristic such as walking gait, touch gestures, keystroke dynamics, behavior profiling, hand waving, power consumption and fusion. A plethora of research works have been published. We present a literature review focusing on the performance of machine learning models. We evaluate each technique separately, but we cannot make a comparison based on performance metrics, as not all researchers use the same metrics. However, for each system, there is at least one of the five basic metrics, namely FAR, TAR, FRR, EER, and Accuracy.

3.1. Walking Gait

Feng et al. [29] investigated the application of two different verification methods the Trajectory Reconstruction and the Statistic Method. They extracted certain features and identified users from the motion

data. Users executed the movement of picking up their phone while sitting/standing stationary or walking. They concluded that user movements (walking) strongly affect the accelerometer data and the verification performance. They used an SVM classifier and achieved 7.09% Equal Error Rate when sitting/standing stationary and 6.13% while walking by the Trajectory Reconstruction and the Statistic Method respectively. They concluded that user movements (walking) strongly affect the accelerometer data and the verification performance.

Muaaz and Mayrhofer [132] used a self-developed android application to collect gait data (from tri-axis accelerometer data) from a smartphone that was placed in the trouser pocket of 35 individuals while they walked with normal pace. To compensate the orientation error produced by the movement of the device during walking, they modified the data processing steps and employed magnitude data, in addition to wavelet-based de-noising modules. Furthermore, they used a modified version of a cycle length estimation algorithm as it was one of the fundamental prerequisites of automatic cycle detection employed in their work. Their method achieved 7.05% EER for a walking session.

Al-Naffakh et al. [30] used the accelerometer and gyroscope of a smartwatch to collect data from 10 individuals while walking during two five-minute sessions on a flat surface on two different days with their natural walking speed. Users also chose on which arm they would wear the smartwatch. They observed that test subjects had different patterns of movement and, thus, walking gait could be used to transparently and continuously authenticate individuals. In an antecedent research using smartwatches, Al-Naffakh et al. [31], presented two experiments on Same, Mixed, and Cross Day situations. Their results showed that the technology is sufficient to discriminate individuals by their gait characteristics. With data from the Same day they achieved 3.12% and 0.13% EER for gyroscope and accelerometer respectively, while in the evaluation of the Cross Day situation they achieved 7.97% and 0.69% for the same sensors. Simultaneously, Kork et al. [101] collected gait data from 50 individuals by using five wearable sensors that were placed to different areas on the body in addition to a smartphone held in hand. Data were collected while users performed slow, normal and fast walking paces. In these three different walking scenarios the Equal Error Rate ranged from 0.17% to 2.27% for the wearable sensors, while for the smartphone the Equal Error Rate ranged from 1.23% to 4.07%.

Shila et al. [102] developed dCASTRA (deep CASTRA), a user verification service which was device-centric and constituted an improved version of a previous service named CASTRA built by the same authors. They extracted gait features for user classification by employing Long Short-Term Memory (LSTM) Recurrent Neural Networks. They also extracted segments of walking by using the Google Activity Recognition Service API. For their experiments, they used data from 15 individuals while they walked from the accelerometer and gyroscope sensors. Their system achieved the identification of users in less than 6 seconds with greater than 99.95% accuracy. In addition, they reported that for learning an efficient model the required amount of data was 60K samples using a 50Hz sampling rate and the minimum duration was 20 minutes.

Baek et al. [159], developed a gait-based authentication framework, named LiSA-G which successfully authenticated and identified 51 users by using the accelerometer and gyroscope sensors of a smartwatch. They applied two scenarios, where, in the first scenario users walked normally towards a smart home system and in the second scenario they walked normally towards a smart car system. Their framework extracted the walking gait and movement of arms and achieved 8.2% EER by using the Random forest classifier, while they needed fewer features and less amount of sensor data, specifically they needed 37 features and 100 data samples respectively, in comparison to other works [95,160,161] for authentication.

Table 2
Walking gait modality research works.

Method	Publications	Platform	Classification	Performance (%)	
				Accuracy	EER
Walking gait	[29] in 2013	smartphone	SVM		6.13
	[132] in 2014	smartphone	Cycle detection		7.05
	[31] in 2017	smartwatch	MLP		0.13
	[101] in 2017	wearable-smartphone	Manhattan method		0.17 1.23
	[102] in 2018	smartphone	LSTM	99.95	
	[159] in 2019	Smartwatch	Random forest		8.2

3.2. Touch gestures

On the issue of user authentication based on touch gestures, Saevanee et al. [32] used finger pressure and keystroke dynamics (hold-time and inter-key) to investigate the behavioral input manner of users. They used a notebook touch pad to collect data and made analysis by the k-NN classification method. The finger pressure achieved 1% EER, the hold-time 30% and the inter-key 35%. Later, Frank et al. [33], used natural navigation gestures on the touchscreen of a smartphone such as up-down and left-right scrolling, to investigate whether these gestures could be used for continuous authentication. They extracted 30 touch features and used a Gaussian RBF kernel support vector machine and a k-nearest neighbor classifier to train user profiles. For inter-session authentication they achieved 2%-3% median equal error rate, while for intra-session authentication they achieved 0%. In the case where they carried out the authentication one week after the enrollment the equal error rate achieved was below 4%. Their method could be implemented to extend screen-lock time or to constitute a part of an authentication system based on multi-modal biometrics.

Later, Li et al. [34] designed a system to continuously re-authenticate smartphones users based on biometrics. They divided participants in 25 target users and in 47 non-target users and selected 8 sliding and tap features. Finger and touch movements of the owner were compared with the current user's finger movement patterns for verification. For classification they used a Support Vector Machine. The highest accuracy achieved was 95.78% for the sliding up gesture. Also, they measured the computation time for their system to make the classification, which was 17 milliseconds. Meanwhile, Zhao et al. [35], used shapes and intensity values of movement and pressure respectively, that were represented by a feature of their own design named Graphic Touch Gesture Feature (GTGF). They used six commonly used touch gestures divided into three datasets. Their method was proven effective since it achieved 2.62% Equal Error Rate when the six gestures were combined. At the same time Serwadda et al. [148], compared 10 classification algorithms based on their performance for touch gestures. They collected data by two self-developed Android applications while users scrolled/swiped back and forth in landscape and portrait orientation of the screen and computed 28 features to represent a stroke. The logistic regression classifier achieved the lowest mean EER of 10.5%, whereas the J48 tree classifier achieved the highest mean EER of 42% across population.

Afterwards, Bo et al. [36] designed SilentSense which was a software-based decision mechanism for smartphones. This mechanism obtained information regarding the usage of applications and the interacting behavior with each application performed by the user from the system API and measured the device's reaction by using the motion sensors. The user was identified as the owner, or a guest based on his actions and the corresponding reaction of the device. The performance of SilentSense was evaluated in walking and static scenarios and

achieved over 99% identification accuracy. Both false rejection (FRR) and false acceptance rates (FAR) were lower than 1% after collecting only 10 actions. In addition, they achieved significantly low error rates for user identification when they combined touch signatures with walking patterns. At the same time, Xu et al. [37] used the keystroke, slide, handwriting and pinch operations on the touchscreen of a smartphone to authenticate users. The best performance achieved by their system was lower than 1 % EER for the slide operation, while for all operations it achieved an EER lower than 10%. Simultaneously, Feng et al. [66] presented an application named TIPS, for the authentication of users based on the location of touch, and the length and curvature of swipe on the touchscreen of three different brands of smartphones. For classification they combined Dynamic Time Warping (DTW) with the One Nearest Neighbor (1NN) classifier and achieved approximately 90% accuracy. Apart from the performance evaluation of their touch-based authentication application they also calculated its power consumption which was 88 mW on average and the battery usage was approximately 6.2%.

Following, Burio et al. [39] built a system that profiled the user based on the his/her finger movements on the touchscreen while signing or writing and the movements of the device. They tested their mechanism on a dataset of 30 volunteers and achieved a True Acceptance Rate (TAR) of 95% with a False Acceptance Rate (FAR) of 3.1% by using a Multilayer Perceptron (MLP). Also, they measured the power consumption and the required time for authentication of their system which was approximately 1000mW and 0.215–0.250s, respectively. Filippov et al. [96] built an authentication system using the data resulting from the interaction of twenty-one users with the touch screen of a smartphone. They gathered 2000 features from seven different gesture types of swipe while users interacted with the smartphone. Their system was evaluated with the use of the Isolation Forest method and achieved values of FRR and FAR equal to 6.4% and 7.5%, respectively. Moreover, their system was able to detect the illegitimate user in seven performed actions.

Shen et al. [104] collected touch data from 102 subjects during three different operation scenarios. More specifically participants hold the smartphone and performed touch actions while sitting, standing still or walking, or they used one hand for the touch operations while the smartphone was placed on a desktop. The best performance among all three operation scenarios was achieved when participants hold the smartphone and performed touch actions while sitting or standing still. More specifically, the FAR was 3.98%, the FRR 5.03%, and the EER 4.71%. They also evaluated the battery, memory, and computation cost of their application and reported that it consumed less than 4.5% of battery per-day, 5.6 Mbytes of memory and it took approximately 110 ms to authenticate a user. Debard et al. [106] proposed the Convolutional Neural Networks, which used 2D filters for the recognition of touch gestures on a touch surface. They used a dataset of 6591 touch

gestures from 27 individuals. They compared the performance of their method to the performance of the methods provided by Hochreiter et al [172] and Liu et al [173] who used the LSTM classifier. The Convolutional Neural Networks classifier used by Debard et al. [106] achieved 89.96% accuracy while the LSTM classifier achieved 73.10% and 87.72% in the works of Hochreiter et al [172] and Liu et al [173], respectively.

Yang et al. [126] used behavioral biometrics of touch and based on anomaly detection they developed a CA method for security-sensitive mobile applications named BehaveSense. Their study was based on the fact that only a few applications contain sensitive data. They used touch gestures of click and slide. They collected 250,000 touch operations in total while 45 participants used the application “WeChat”. For classification they used Isolation Forest and One-class SVM Method. For the sequence of touch operation their method achieved an average accuracy of 95.85% when considering 9 touch operations. They also calculated the energy consumption of their method which was 6.82 mAh, while the average calculating time was less than 0.01 s.

Alqarni et al. [181] used the accelerometer and gyroscope sensors of a smartphone to collect data resulting from keystroke dynamics, touch gestures, and hand waving. They divided their data into two groups, namely short-term activities and gestures. Data from the short-term activities included the data collected while participants were using a smartphone and executed specific typing and hand waving tasks. Data from gestures included data that were collected while participants were holding the smartphone in their hand and performed predefined swiping, drawing, and hand waving gestures. They evaluated their method by using 3 different classifiers, namely SVM, Random Forests, and Bayes Net. Regarding the group of short-term activities, the best performance was achieved while participants typed a predefined sentence. More specifically, they achieved 76.20% accuracy, and 0.133 RMSE, by using the Random Forest classifier. Regarding the group of hand gestures, the best performance was achieved while participants performed two different gestures. The first gesture was the drawing of a circle with their right hand and the second was the drawing of a triangle with their right hand. In the first gesture, they achieved 92.81% accuracy, and 0.1218 RMSE, while in the second gesture, they achieved 92.81% accuracy, and 0.1178 RMSE. In both gestures, they used Random Forest for classification.

3.3. Keystroke dynamics

Most techniques are based on a specific context with predefined text. Clark and Furnell [40] authenticated users based on the user’s patterns of typing when entering text messages and telephone numbers. By using Multi-Layer Perceptron (MLP), Radial Basis Functions (RBF) and General Regression Neural Network (GRNN) classifiers, they achieved an average EER of 12.8%.

In another study, Feng et al. [41] asked participants to enter 4-digit passwords during the log-in session and a predefined sentence of 20 characters during the post-login session. The keystrokes were collected by an Android application. The authentication performance during the log-in session achieved an FRR approximately 11.0% using Bayes Net. In the post-login stage, the Random Forest algorithm outperformed the other two (8.93% vs 19.78% and 14.26%FAR).

Simultaneously, Draffin et al. [44] collected keystrokes from 13 participants in a real-world study in a three weeks period. They did not intervene and collected 86000 keystrokes not restricted to passwords or predefined sentences. By using Feed Forward Neural Network for classification they achieved 86% accuracy after 15 keypresses with 14% FAR and 2.2% FRR.

Later, Buschek et al. [42] asked 28 participants to type 20 times 6 different passwords in non-specific order and in three different hand postures. By using the Least Squares Anomaly Detection, they achieved 17% EER when combining pressure with spatio-temporal features. Following, Zhang et al. [110] collected time and pressure-related data from 10 students while they typed the same password by using a self-prepared Android App. Their user authentication model was based on an RBF network and achieved 10.3% false positive rate and 91.7% true positive rate.

Darren and Inganez [111] collected data from the input of 15 predefined sentences during four different typing scenarios. Participants could choose one of the four scenarios: One Handed Stationary, Two handed Stationary, One Handed Moving, Two handed Moving. All 4 activities had to be completed by the smartphone owner. All their results, using a Least Squares SVM classifier with RBF kernel, achieved around 1% EER, with one-handed scenario being the best at 0.44% EER, 100% accuracy, 0% FAR and 1 % FRR Krishnamoorthy [43] classified users based on keystroke dynamics, by applying machine learning concepts. Participants typed a specific password to record their typing characteristics. She effectively identified each of 94 users who participated achieving an identification accuracy of 98.44% when using the random forest classifier.

Table 3
Touch gestures modality research works.

Method	Publications	Platform	Classification	Performance (%)		Accuracy	FRR	EER	RMSE
				FAR	TAR				
Touch gestures	[32] in 2008	smartphone	k-NN			99		1	
	[33] in 2012	smartphone	k-NN & SVM					<4	
	[34] in 2013	smartphone	SVM			>88.28			
	[35] in 2013	smartphone	L1 distance					2.62	
	[148] in 2013	smartphone	logistic regression					10.5	
	[36] in 2014	smartphone	SVM	<1		99	<1		
	[37] in 2014	smartphone	SVM					<1	
	[66] in 2014	smartphone	Combination of DTW with 1NN			90%			
	[39] in 2016	smartphone	MLP	3.1	95				
	[96] in 2018	smartphone	Isolation Forest	7.5			6.4		
	[104] in 2018	smartphones	HMM	3.98			5.03	4.71	
	[106] in 2018	touch surface	CNNs			89.96			
	[126] in 2019	smartphone	Isolation Forest			95.85			
	[181] in 2020	smartphone	Random Forest			92.81			0.1178

Table 4

Keystroke dynamics modality research works.

Method	Publications	Platform	Classification	Performance (%)		Accuracy	FRR	EER
				FAR	TAR			
Keystroke Dynamics	[40] in 2006	smartphone	MLP, RBF, GRNNs					12.8
	[41] in 2013	smartphone	RF	8.93				
			J48	19.78				
			Bayes	14.26			11.0	
	[44] in 2013	smartphone	FFNN	14		86.0	2.2	
	[42] in 2015	smartphone	LSAD					17.00
	[110] in 2016	smartphone	RBF		91.7		10.3	
	[111] in 2018	smartphone	SVM	0		100	1	0.44
	[43] in 2018	smartphone	RF			98.44		2.2

3.4. Behavioral profiling

Neal et al. [50] developed a method to profile users based on Bluetooth sightings, Wi-Fi access points and application usage. To produce matching scores, they utilized a categorical nearest-neighbor classifier. They achieved 80% identification rate when using Bluetooth, 93% when using Wi-Fi, 77% when using applications and 85% when combining the features. Following, Anjomshoa et al. [115] presented TrackMaison, a mobile application tool, which kept track of the manner smartphone users made use of five social network services via the usage of data, the usage frequency, the location and session duration. By using their tool, they conducted a study on several user profiles in Instagram and showed that users which are highly active can be identified with 3% FAR.

Fridman et al. [118] suggested an authentication method that regards the user's application and website logs, the keystroke dynamics and the location of the device as defined from the Wi-Fi or the GPS. Their data collection involved behavioral biometrics from 200 participants who used their smartphone for 30 days. They used the support vector machines (SVMs) and the equal error rate (ERR) achieved by their authentication system was 1% after 30 minutes of interaction and 5% after 1 minute.

Later, Mahbub et al. [119] used unique application usage data and handling the unexpected events in the data. They collected data for approximately six months, from 218 participants. These data included the device location, the installed, removed or updated applications, the currently running foreground application, etc. They divided their dataset into three categories, namely: “*all observations*” category, “*all except the ones with unknown applications*” category and “*all without unforeseen observations*” category. The number of training samples that they used were 500. By using the M-ED algorithm their method achieved an EER of 34.31% in the “*all observations*” category, 42.47% in the “*all except the ones with unknown applications*” category and 43.29% in the

“*all without unforeseen observations*” category. Moreover, their method detected an intrusion in 2.5 minutes of application use.

Alotaibi et al. [162] employed the usage patterns of specific applications to identify and verify users on mobile phones. More specifically, they collected data from 100 individuals while they used their device as usual, and employed the applications of Phone Call, SMS, Download, YouTube, WhatsApp, Browser, Google Play, Email, Viber, Google Photo, Camera, and Yahoo mail. Each individual participated in the data collection procedure for one month. Their approach succeeded in successfully identifying users by their behavioral profiling, with 26.98% average EER and the Gradient Boosting Classifier.

Pang et al. [180] developed an application to collect data from seven modalities from a smartphone, namely, LOCATION, ACTIVITY, APP, BLUETOOTH, WiFi, CALL, and SMS. They evaluated the performance of their approach by using different classifiers. The best results were achieved with the MineAuth method which was 98.5% accuracy. The authors also measured the running time, the power consumption, the CPU, and memory usage of their approach. They reported that the total running time was 90.601 ms. More specifically, the running time for behavior construction was 87,759 ms, for habits mining and authenticator it was 2310 ms, while for the decisionmaker it was 532 ms. The total power consumption was 0.37%. More specifically, the power consumption for behavior construction was 35% MB, for habits mining and authenticator it was 0.01%, while for the decisionmaker it was 0.01%. The total CPU usage was 46.85%. More specifically, the power consumption for behavior construction was 15.30%, for habits mining and authenticator it was 16.10%, while for the decisionmaker it was 15.45%. The total memory usage was 186,29MB. More specifically, the memory usage for behavior construction was 75.10 MB, for habits mining and authenticator it was 60.2 MB, while for the decisionmaker it was 50.99 MB.

Table 5

Behavioral profiling modality research works.

Method	Publications	Platform	Classification	Performance (%)		EER
				FAR	Accuracy	
Behavioral profiling	[50] in 2015	smartphone	Categorical Nearest-Neighbor		93	
	[115] in 2016	smartphone	KNN	3		
	[118] in 2017	smartphone	SVMs			1
	[119] in 2018	smartphone	M-ED algorithm			34.31
	[162] in 2019	smartphone	Gradient Boosting			26.98
	[180] in 2019	smartphone	MineAuth method		98.5	

3.5. Hand waving

Sitová et al. [38] used Hand Movement, Orientation, and Grasp (HMOG) to continuously authenticate smartphone users. They used readings from the accelerometer, gyroscope, and magnetometer sensors to capture orientation features and subtle micro-movement as the user grasps, holds, and taps on the smartphone. They evaluated their mechanism by collecting data under both walking and sitting circumstances. They achieved 7.16% EER when walking and 10.05% when sitting by combining HMOG, keystroke features and tap. Also, they measured the energy consumed by their system to compute the 18 HMOG features from the magnetometer, accelerometer and gyroscope sensor readings which was 0.08 joules. Yang et al. [100] employed the waving patterns to lock and unlock the smartphone in a system named OpenSesame. They collected data from 200 subjects and used an SVM classifier. The result from their experiment achieved 15% mean FAR while the FRR < 8%. Later, Buriro et al. [98] developed an authentication mechanism by using the micromovements of the user's hand(s) within a few seconds after unlocking his smartphone. They collected data from the smartphone sensors and more specifically from the gyroscope, the accelerometer, the gravity, the orientation and the magnetometer. On a dataset of 31 volunteers their mechanism achieved a 96% TAR at 4% EER using the Random Forest classifier.

3.6. Power consumption

Zhang et al. [15] presented Power Tutor, a power consumption monitoring model on electronic devices which utilized the battery sensors. The absolute average error rate achieved by their model was lower than 10%. Murmura et al. [130] showed that the operation state of a smartphone's device driver affects its power consumption. They succeeded in attributing power consumption to applications with lower than 4% error rate by using the Ordinary Least Squares regression analysis. More specifically for the applications Logger, Music, Sound Recorder and GPSTracker the EERs were 3.69 %, 1.27 %, 0.74 % and 3.68% respectively.

Shye et al. [129] presented a model to estimate power that was tested on 20 users for more than a week. They used a self-developed logger application to record the activity of users and showed that power consumption patterns are differentiating greatly among users. They found that this approach was not effective since they could not profile users on the basis of power consumption patterns.

Murmura et al. [86] used a machine learning based approach that involved three different modalities: power consumption, physical movement and, touch gestures to continuously monitor users. They collected data from 73 participants while they used facebook and chrome. They applied the Strangeness-based Outlier Detection algorithm for each modality and achieved an EER that ranged from 6.1% to 6.9% depending on training times that varied between 20 and 60 minutes and sufficient data. They deployed their approach and created the kryptowire system which is available on the market [51]. This application employs the embedded sensors of mobile devices including power consumption, physical movement and, touch gestures to identify users based on the way they interact with their device and within the context of each mobile application.

3.7. Fusion

Saevanee et al. [24] conducted a study where they first employed behavior profiling, keystroke dynamics and linguistic profiling as single modalities to discriminate users and achieved error rates of 20%, 20% and 22% respectively. Following, they combined behavior and linguistic profiling with keystroke dynamics by applying matching-level fusion and increased the reliability of continuous authentication systems with an overall EER of 8% with MLP classifier.

Later, Zheng et al. [52] combined four features for a PIN typing action on a smartphone touch screen (acceleration, pressure, size, and time). They collected tapping data from more than 80 users and they first evaluated the performance of the modalities as single modalities and following they measured the results of fusing these modalities. The fusion achieved better results in comparison to the single modalities results. More specifically, the size modality, as a single modality achieved 27% EER, the pressure modality achieved 16% EER, the time modality achieved 14% EER, and the acceleration modality achieved 10% EER. When they combined the modalities of size, time and acceleration they achieved 3.65% EER with k-NN classifier.

Buriro et al. [53] used features collected from 26 participants while they unlocked a smartphone before answering a call. They collected features of slide swipe, the manner of moving the arm while carrying the device towards the ear and recognition of voice. They first evaluated the modalities as single modalities and achieved 22.28% FAR and 4.84% FRR for the slide modality, 26.69% FAR and 6.19% FRR for the pickup modality and 63.92% FAR and 12.69% FRR for the voice modality. In their multimodal system they incorporated the features of slide and pickup and enhanced the unimodal result, with 11.01% FAR and 4.12% FRR while the final HTER was 7.57% using the BayesNet classifier. The computation time required for the same classifier was 64, 762 and 205 ms for the slide, pick up and voice modalities respectively.

At the same time, Haq et al. [121] applied micro-environment sensing to build a three-class smartphone user classification scheme based on physical activity recognition. They employed time and frequency domain features to recognize physical activities such as walking, running, etc. They used a probabilistic scoring model to recognize the activity patterns and subsequently classify users as authenticated, restricted access user or guest user (supplementary) and impostor. Their scheme granted full access to authenticated users, restricted access to supplementary users and zero access to impostors. For their experiments they used a dataset created by Shoaib et al. [120,123]. To enhance their authentication scheme, they took into account the work of Yang et al. [124] regarding the smartphone's position sensitivity issue and incorporated micro-environment sensing, meaning the awareness of the smartphone's close surroundings. For this purpose, they selected five different body positions to place the smartphone while users performed physical activities. The body positions were the wrist, the upper arm, the waist, the left and right thigh, and were considered as the smartphone's close surroundings. Their authentication model was trained not only to sense the smartphone's position on the human body but to also combine it with the performed physical activities of each user. First they evaluated the modalities as single modalities and reported that they achieved an average accuracy of 97.55% when the smartphone was placed to the waist, 98.57% when the smartphone was placed to left thigh, 98.01% when the smartphone was placed to right thigh, 95.45% when the

Table 6
Hand waving modality research works.

Method	Publications	Platform	Classification	Performance (%)			
				FAR	TAR	FRR	EER
Hand waving	[38] in 2015	smartphone	SVM	15	96	8	7.16
	[100] in 2015	smartphone	SVM				
	[98] in 2017	smartphone	Random Forest				4

smartphone was placed to the upper arm, and 96.85% when the smartphone was placed to the wrist. Following, they combined data from the accelerometer, gyroscope and magnetometer sensors and achieved an average accuracy rate of 97.38% by the Bayes Net classifier where the average error rate was 0.027 MAE and 0.086 RMSE. They also reported that the best TAR achieved was 0.95 for the impostor with the smartphone placed in his waist and the best FAR was 0.03 and achieved for the supplementary user when the smartphone was placed on his waist and his left thigh. They also measured the energy and storage cost of their application and reported that it consumed between 5% and 11.2% of energy depending on how frequently the users were moving and maximum 1.8MB of storage per day. In addition, they measured the required time for the BayesNet classifier to perform authentication and reported that it 5.61 s.

Kumar et al. [134] investigated if arm movements while walking can be used for user authentication. In their experiments, they used the accelerometer and the gyroscope sensors of a smartwatch to extract 32 acceleration and 44 rotation features respectively. They used the rotation and acceleration of arms and their fusion at the feature level to build three different continuous authentication designs. They implemented these designs by four classification algorithms resulting in twelve authentication mechanisms and tested these mechanisms in three different environments where participants wore the smartwatch on their wrist and walked naturally. They first evaluated the performance of the modalities as single modalities and achieved a mean dynamic FAR of 2.6% and a dynamic FRR of 4.3% for the acceleration modality and 4.2% dynamic FAR and 10.5% dynamic FRR for the rotation modality. Following they deployed score level fusion at the feature level with the k-NN classifier and the best results achieved were 2.2% dynamic FAR and 4.2% dynamic FRR in the inter-phase environment. The authors also reported the number of features vectors that were in the training data set which was 156 for both the authentic and impostor users.

Shrestha et al. [64], applied a multi-modal walking biometrics approach, namely Walk Unlock ZIA(WUZIA) on a Zero-Interaction Authentication (ZIA) system. ZIA is a paradigm where a legitimate user carries a prover device and walks towards a verifier device which unlocks automatically [49]. They used the accelerometer, gyroscope and magnetometer of a smartphone and a smartwatch to collect data from 18 individuals. The smartphone was placed in their pocket to record hip movement and the smartwatch was worn on their wrist to record hand movement. Participants walked towards the verifier device which would unlock and respond to the prover device only after it detected the legitimate user's walking patterns. First, they evaluated the performance of the watch which achieved 0.46% FRR and 0.63% FAR and the phone which achieved 0.18% FRR and 0.36% FAR. Following they combined the watch with the watch and achieved the almost error free FRR of 0.2% and FAR of 0.3% on average with the Random Forest classifier. Their results showed that their approach detected with high accuracy a legitimate or a non-legitimate user when using both of the aforementioned devices. ZIA systems are already deployed in several real-world applications like BlueProximity [174], Keyless-Go [175], PhoneAuth [176], and access control systems based on wearable devices like the iWatch [177]. BlueProximity [174] enables the unlocking of a computer screen when the user approaches the computer and holds a mobile device. Keyless-Go [175], allows users to start and lock their car just by having the key with them. PhoneAuth [176], is deployed based on the scenario where users, for example, walk towards an Internet kiosk to navigate to a web page and they type their username and password to log in. Following, PhoneAuth stores the cryptographic credentials on the user's phone. If the phone is present when the user logs into a site, then it will ascertain the user's identity via Bluetooth with the computer's browser. The iWatch [177] when it is in close range to iPhones and Macs it allows the legitimate user to log in without using a pin or a password. Hence, WUZIA can be implemented in a traditional ZIA system such as BlueProximity [174], Keyless-Go [175], PhoneAuth [176], and the iWatch [177]. More specifically WUZIA can be implemented in

BlueProximity [174] simply by changing the app and without changing the terminal software. WUZIA can also be implemented on systems like Keyless-Go [175] where the key has incorporated RF capability and processor.

Li and Bours [122] developed a mobile application authentication method in which they collected and analyzed data from four resources: gyroscope and accelerometer sensors, Bluetooth, and Wi-Fi. With the scores from the aforementioned resources they performed a score-level fusion. The 321 participants who joined the experiment were free to choose when and where they would open the App while performing seven different activities and carrying the smartphone at a different position. For the gyroscope features the EER achieved was 28.12% while for the accelerometer features the EER achieved was 36.11%. The best EER achieved by their method was 9.67% for the score-level fusion with data from all four resources and Random Forest classifier.

Later, Buriro et al. [125] built an authentication mechanism named AnswerAuth that was based on the behavioral biometrics of smartphone users. More specifically, AnswerAuth utilized the manner users slide the screen's lock button to unlock the smartphone and bring it close to their ear. Data were recorded by using the incorporated sensors of the smartphone, i.e., touchscreen, accelerometer, gyroscope, gravity and magnetometer. Their dataset derived from 85 individuals while they unlocked the smartphone in three different positions, namely walking, standing, and sitting. They reported that the best results achieved were the fusion of the slide operation with the bringing the device towards to the ear while users were standing. Specifically, they achieved a TAR of 99.35% and 98.98% accuracy by using the Random Forest (RF) classifier.

At the same time, Volaka et al. [178] used the touch-screen data (scroll), and data from the accelerometer and the gyroscope sensors from the HMOG dataset, as provided in the work of Sitova et al. [38]. First, they evaluated only the touch-screen data using the LSTM classifier and achieved an average accuracy of 88%, an average TAR of 78-79%, and an EER of 16%. Following, they evaluated the combination of touch-screen data and data from the accelerometer and achieved an average accuracy of 89%, an average TAR of 79%, and an EER of 16%. Next, they evaluated the combination of touch-screen data and data from the gyroscope and achieved an average accuracy of 89-90%, average TAR of 80-81%, and an EER of 14%. Finally, they evaluated the combination of touch-screen data and data from the gyroscope and the accelerometer and achieved an average accuracy of 88%, an average TAR of 78%, and an EER of 15%.

Simultaneously, Lamiche et al. [179] used the gait patterns resulting from the accelerometer and the keystroke dynamics while 20 users walked and typed a predefined sentence. They evaluated their method under four different scenarios, by using different classifiers, and by applying a feature level fusion method to profile users by using both modalities of walking gait and keystroke dynamics. In the first scenario, users walked naturally while the smartphone was placed in the pocket of their trouser and achieved 96.54% accuracy and 9% EER with the Random Forest classifier. During the second scenario, users walked naturally while they were holding the device on their hand and they achieved 97.34% accuracy and 6% EER by using the Random Forest classifier. In the third scenario, users answered a phone call while they were walking and achieved 98.1% accuracy and 0.3% EER with the Random Forest classifier. During the fourth scenario, users typed a predefined sentence while they were walking and they achieved 99.11% accuracy, 0.684% average FAR, 7% FRR, and 1% EER by using the Multilayer Perceptron (MLP) classifier.

Following, Abuhamad et al. [182] presented a continuous authentication method based on deep learning, named AUtoSen. The performance of their method was evaluated by using LSTM on four different sets of data. The data collected included touch screen data such as sliding or tapping while participants interacted with their smartphones, and sensor data from the accelerometer, the gyroscope, the magnetometer, and elevation, while the individuals performed various

Table 7
Power Consumption modality research works.

Method	Publications	Platform	Classification	Performance (%) EER
Power Consumption	[15] in 2010	smartphone	PowerTutor model	>10
	[130] in 2012	smartphone	OLS	>4
	[86] in 2015	smartphone	StrOUD	6.1

physical activities. These data were divided into four groups. The first group included data from the accelerometer, the gyroscope, the magnetometer, elevation, and touch screen data. The second group included data from the accelerometer, the gyroscope, the magnetometer, and elevation. The third group included data from the accelerometer, the gyroscope, and the magnetometer, while the fourth group included data from the accelerometer and the gyroscope. The authors showed that their method achieved 0.41% EER, 0.95% FAR, and 6.67% FRR, by using sensory data of one second, from only three sensors, namely the accelerometer, the gyroscope, and the magnetometer.

3.8. Discussion on behavioral biometrics

In this section, we make a discussion on the machine learning and deep learning models that we previously presented. Unfortunately, we cannot compare the performance of machine learning algorithms given in the literature because researchers use different evaluation criteria and different kinds of data to evaluate machine learning algorithms. However, we report the best results that were achieved in each category. Moreover, we summarize some advantages and disadvantages of the Methods as shown in Table 9. A discussion on each modality follows:

- **Gait:** As we saw above the walking gait modality can be efficiently used to transparently and continuously authenticate individuals as this modality sufficiently discriminates different users by their gait characteristics [31]. An accuracy of 99.95% was achieved by using the Long Short-Term Memory (LSTM) Recurrent Neural Networks classifier [102]; while an EER of 0.17% and 0.13% was achieved by applying the Manhattan method [101], and the MLP classifier [31] respectively. In addition, the walking gait modality can work without user intervention or additional hardware apart from the smartphone embedded components. However, movements of users while walking, for example the MDP motion (the movement of picking up their phone to the ear), strongly affects the accelerometer data and the verification performance [29]. This is due to Einstein's principle of equivalence [45] according to which the acceleration data incorporates the effect of motion acceleration as well as the gravity acceleration. To solve this, Feng et al. [29] removed the accelerometer data and emphasized on gyroscope and magnetometer data and increased the performance result. An additional disadvantage is that the accuracy can be negatively affected due to the uneven ground, possible injuries, footwear, fatigue, personal peculiarities, etc. [86]. As highlighted in the work of Shila et al. [106] machine learning techniques are used in the design of walking biometrics and current methods are either based on feature extraction for model training or they cannot learn the time series data. Thus, they constitute a major challenge, leading to an increased number of false positives and false negatives. To extract temporal features and successfully classify users, they used a deep learning architecture, namely LSTM (Long Short Term Memory) Recurrent Neural Networks. In the same work, they also compared the performance of LSTMs to CNNs and reported that while both achieved an excellent performance, the latter are more difficult to train since they required four hours while the former only thirty minutes.
- **Touch gestures:** The touch gestures modality combines several touch operation patterns and achieves very good results. The k-NN and the SVM classifier achieved 99% accuracy in the works of [32] and [36] respectively, while SVM and Isolation forest classifiers achieved an

average accuracy of 95.85 [126]. Subsequently, touch biometrics hold much promise as a method for the passive and continuous authentication of individuals [37]. Moreover, for the collection of data no additional equipment is required except for the incorporated device sensors. When an individual performs touch gestures the dynamic nature of this action results in a test sample that can in part noticeably vary from prior training samples of the same individual. For this reason, an authentication system might reject the testing sample, in case it expects an individual to perform the gestures in the same manner [137]. A dataset that contains millions of samples is necessary to realistically evaluate the performance of an authentication system based on touch gestures [11]. So far, up to our knowledge, the biggest sample comprises of 250,000 touch operations [126]. Debard et al [102] mentioned the issue of sequential data classification in touch gestures that present different sums of data per time. They used Convolutional Neural Networks (CNN) and improved the recognition rate in comparison with the commonly used Recurrent Networks. More specifically, Debard et al. [106] achieved 89.96% accuracy while the LSTM classifier achieved 73.10% and 87.72% as reported in the works of Hochreiter et al [172] and Liu et al [173], respectively.

- **Keystroke:** The authentication of users with the use of keystroke dynamics modality requires no additional equipment apart from the built-in sensors of the device. In addition, it is non-intrusive and efficient in capturing the users' patterns without disturbing them. Moreover, the results of the authentication of users with the use of keystroke dynamics modality are of high accuracy. The SVM and the Random forest classifier achieved an EER of 0.44% and 2.2% in the works of [111] and [43] respectively; while the RBF classifier achieved 91.7% TAR [110]. A disadvantage of an authentication system based on keystroke dynamics is the inconsistent performance if users perform inconsistently.
- **Behavioral profiling:** The behavioral profiling approach is based on the assumption that the majority of users tend to perform similar tasks due to their habitual nature [77]. Thus, in case users interact inconsistently with their mobile phones results in unreliable performance which constitutes a major deficiency [11]. Nevertheless, the results of the authentication of users with the use of behavioral profiling are of high accuracy. The SVM classifiers achieved 1% EER in the work of [118], while in the work of [180] the MineAuth method achieved 98.5% accuracy. The KNN achieved 3% FAR. The CNN achieved 93% accuracy and the M-ED algorithm achieved an EER of 34.31%. Finally, this method does not require any additional hardware apart from the embedded smartphone components.
- **Hand-waving:** Regarding the hand-waving modality very little research is conducted. The random forest classifier has achieved 4% EER in [98]. This approach does not involve any user interaction. For the collection of motion data, no additional equipment is necessary except for the incorporated device sensors.
- **Power consumption:** The power consumption modality cannot be used to efficiently profile users as a single modality [129], but only in combination with other behavioral biometric modalities [86]. The measurement of power consumption resulting from the user's activities does not require any additional equipment apart from the incorporated sensors of the device's battery driver [15]. An EER smaller the 4% was achieved by using the OLS classification [130].
- **Fusion:** A plethora of combinations is used for the authentication of individuals with the fusion of biometrics, which achieves improved

Table 8

Fusion research works.

Method	Publications	Platform	Classification	Performance (%)							
				FAR	TAR	Accuracy	FRR	EER	MAE	RMSE	HTER
Fusion	[24] in 2011	Smart phone	MLP					8			
	[52] in 2014	Smart phone	k-NN					3.65			
	[53] in 2015	Smart phone	BayesNet	11.01			4.12				7.57
	[134] in 2016	Smart watch	k-NN	DFAR 2.2			DFRR 4.2				
	[64] in 2016	Smart phone& watch	Rand. Forest	0.3			0.2				
	[121] in 2017	Smart phone	Bayes Net	0.03	0.95	97.38			0.027	0.086	
	[122] in 2018	Smart phone	Random forest					9.67			
	[125] in 2019	Smart phone	Random Forest		99.35	98.98					
	[178] in 2019	Smart phone	LSTM		78	88		15			
	[179] in 2019	Smart phone	MLP	0.684		99.11	7	1			
	[182] in 2020	Smart phone	LSTM	0.95			6.67	0.41			

results compared to single modality methods [24,52,53,64,122,134]. For example, in [24] the behavior profiling, the keystroke dynamics and the linguistic profiling as single modalities achieved error rates of 20%, 20% and 22% respectively. When combined by applying matching-level fusion the reliability of the system increased with an overall EER of 8%, with MLP classifier. In [52] the modalities of size, pressure, time and acceleration achieved EERs of 27%, 16%, 14%, and 10% respectively as single modalities and when they were combined, the EER was 3.65% with k-NN classifier. In the work of [53] when they evaluated the modalities as single modalities, they achieved 22.28% FAR and 4.84% FRR for the slide modality, 26.69% FAR and 6.19% FRR for the pickup modality and 63.92% FAR and 12.69% FRR for the voice modality. When they incorporated the features of slide and pickup into their multimodal system, they enhanced the unimodal result, with 11.01% FAR and 4.12% FRR using the BayesNet classifier. In the work of [134] the performance of the modalities as single modalities achieved a mean dynamic FAR of 2.6% and a dynamic FRR of 4.3% for the acceleration modality and 4.2% dynamic FAR and 10.5% dynamic FRR for the rotation modality. When the authors deployed score level fusion at the feature level with the k-NN classifier they achieved 2.2% dynamic FAR and 4.2% dynamic FRR in the inter-phase environment. In the work of [179] they achieved 99.11% accuracy, 0.684% average FAR, 7% FRR, and 1% EER by using the Multilayer Perceptron (MLP) classifier, while users were walking and typed a predefined sentence. In the work of [64] the evaluation of the performance of the watch achieved 0.46% FRR and 0.63% FAR, while the phone achieved 0.18% FRR and 0.36 % FAR. When they combined the watch with the watch, they achieved the almost error free FRR of 0.2% and FAR of 0.3% on average with the Random Forest classifier. In the work of [122] the gyroscope and the accelerometer features achieved EERs of 28.12% and 36.11%, respectively. When the author applied score-level fusion with data from all four resources and Random Forest classifier they achieved 9.67% EER.

The only work where a single modality achieves better results in comparison with fusion results is the work of Haq et al. [121]. Specifically, when the authors evaluated the modalities as single modalities they reported that average accuracy achieved was 97.55% when the smartphone was placed to the waist, 98.57% when the smartphone was placed to left thigh, 98.01% when the smartphone was placed to right thigh, 95.45% when the smartphone was placed to the upper arm, and 96.85% when the smartphone was placed to the wrist. When they combined data from the accelerometer, gyroscope and magnetometer sensors they achieved an average accuracy rate of 97.38% by the Bayes Net classifier. As we can see, the single modalities of the smartphone being to the upper arm or when it is placed to the wrist achieved better results. The random forest achieved 99.35% TAR in [125]. The random forest classifier also achieved very good results in the deployment of WUZIA [64] scoring 0.2% FRR and 0.3% FAR on average. A drawback of the WUZIA approach is that it incorporated the use of a smartwatch as an additional token. The collection of data requires no additional equipment apart from the built-in sensors of the device. Compared to unimodal biometric systems these systems are evoking interest regarding their implementation in large scale authentication systems even though they involve more computational cost, they take higher processing time and require more storage [138,139].

As we saw, most of the works presented in this paper are using the zero-effort evaluation approaches to test the performance of their authentication systems. The zero-effort attack is based on the sufficient similarity of the templates between the attacker and the legitimate user and relates to the uniqueness property of a biometric characteristic. It does not involve any sophisticated action by the adversary. An adversary attack involves sophisticated action by the attacker to successfully impersonate a legitimate user. The level of sophistication strongly depends on the available resources such as digital or physical means, time, and information regarding the biometric system and the victim [95]. We believe that there is a need to shift from the zero-effort to high-effort evaluation approaches to see the actual performance of the systems under the spectrum of today's possible threats. In addition, user

Table 9

Advantages and disadvantages of each method.

Methods	Advantages	Disadvantages
Walking Gait	Continuous Authentication. Can work without user intervention. Doesn't need additional hardware.	The verification performance is greatly affected by Mobile Device Picking-up (MDP) motion of users. Accuracy can be negatively affected due to uneven ground, possible injuries, footwear, fatigue, personal peculiarities, etc.
Touch Gesture	Continuous Authentication. A promising method. Doesn't need additional hardware.	Noticeable variations in the testing sample Research is conducted on small-scale samples.
Keystroke dynamics	Continuous Authentication. Doesn't need additional hardware. Doesn't disturb users. High accuracy of results.	Doesn't work well if users perform inconsistently.
Behavioral profiling	Continuous Authentication. Doesn't need additional hardware.	Doesn't perform well if users perform inconsistently.
Hand waving	Continuous Authentication. Does not require any interaction. Doesn't need additional hardware.	Very little research is conducted.
Power consumption	Continuous Authentication. Doesn't need additional hardware.	Cannot profile users solely on this basis.
Fusion	Continuous Authentication. Achieves improved results. Doesn't need additional hardware.	Require more storage. More processing time. Higher computational and power cost.

behaviors and habits may change over time, thus authentication systems should be able to adapt to these changes. A paradigm that could be employed towards this direction is Continual Learning [163,167,168]. This framework can train successfully deep discriminative models as well as deep generative models in complex continual learning scenarios where new tasks appear, and existing tasks change over time.

Also, most researchers employ commonly used machine learning techniques, and only a few used deep machine learning. Debar et al. [106] employed CNNs in the category of touch gestures and Shila et al. [102] used LSTMs and CNNs in the category of walking gait. Therefore, it is impossible to make a comparison between them since they are only a few and they belong to different categories of behavioral biometrics. In the work of Shila et al. [102] they showed that LSTMs are faster to train than CNNs but they both performed excellently.

Finally, there are several additional aspects of machine learning on mobile devices that should be taken into consideration when evaluating an authentication approach. These aspects include the amount of data from each user that is necessary for each approach to be effective, the computational and communication cost, and the usage of battery, memory, and CPU. However, most researchers do not measure these aspects in their works. Still, we are presenting all the research work that measure these aspects in their results in Table 10 that follows. We believe that future research should measure those aspects to enable a comparison of each method based on these aspects. Moreover, we believe that researchers should at least weight and report the FAR, TAR, FRR, EER, and Accuracy when evaluating the performance of their approach. Also, they should measure and report additional aspects of machine learning such as the amount of data from each user that is necessary for each approach to be effective, the computational and communication cost, and the usage of battery, memory, and CPU. In this way, it will be feasible to make a comparison between the approaches.

4. Possible attack vectors on BBCA systems

As we mentioned previously, we believe that there is a need to shift from the zero-effort to high-effort evaluation approaches to see the actual performance of machine learning and deep learning models under the spectrum of today's possible threats. In this section we review possible attack vectors on CA systems.

4.1. Practical attack techniques

A behavioral biometric system could face several threats such as

malware and a form of shoulder surfing attacks, mimic, impersonation, spoofing, replay, statistical, algorithmic, robotics, etc. These attacks can be divided into two general categories zero-effort or passive attacks and adversary or active attacks. The zero-effort attack does not involve any sophisticated action by the adversary. It is based on the sufficient similarity of the templates between the attacker and the legitimate user and relates to the uniqueness property of a biometric characteristic. An adversary attack involves sophisticated action by the attacker to successfully impersonate a legitimate user. The sophistication level of an adversary attack strongly depends on the available resources such as digital or physical means, time, and information regarding the biometric system and the victim [95].

Biometric researchers focus more on mimicry attacks since they do not involve any modification of the authentication system or the device and can be launched easily and unnoticeably on any biometric-based authentication system. Up to our knowledge, three kinds of mimicry attacks can be launched in authentication systems that are based on behavioral biometrics namely, the zero-effort, the minimal-effort, and the high-effort mimicry attacks. In a zero-effort mimicry attack, imitators are randomly selected by the attackers either from or outside the database since they do not possess any information regarding the legitimate user. In minimal-effort mimicry attack, imitators are selected by the attackers according to specific criteria since they possess information regarding the legitimate user. In high-effort mimicry attacks, imitators are intensively trained to successfully mimic the legitimate user [134]. Following we will see practical attacks and threats on walking gait, keystrokes and touch gestures.

4.2. Attacks on walking gait

A number of studies examine the vulnerability of authentication based on the gait patterns of an individual. The studies presented here refer to passive zero effort impostor attacks, active impostor mimic attacks including minimal effort impersonation attacks [142] and attacks after training of the adversary [60,64,95]. The aforementioned attacks can be divided in two general categories passive or zero-effort attacks and active or adversary attacks. In an active attack the attackers try to be accepted by the authentication system. They use two ways either by changing their biometric in order to match another targeted person or by selecting a victim based on information, for example the closest in the database, to verify themselves against the template of this targeted person. Apparently, the attackers can combine both ways.

Gafurov et al. [142] set two different scenarios, namely a friendly

Table 10
Research works that measured additional aspects of machine learning.

Method	Publications	Classification	Performance Computation time for authentication	Memory usage	CPU usage	Battery consumption	Necessary amount of data for training	Necessary duration of training	Necessary amount of features for authentication	Necessary amount of data samples for authentication
Walking gait	[102] in 2018 [159] in 2019	LSTM Random forest					60K samples	20 minutes	37	100
Touch gestures	[34] in 2013 [66] in 2014 [39] in 2016 [104] in 2018 [126] in 2019	SVM DTW with 1NN MLP HMM Isolation Forest	17 ms 0.215–0.250 s 110 ms 0.01 s	5.6 Mb		88 mW on average $\approx 6.2\%$ 1000mW 4.5% 6.82 mAh				
Behavioral profiling	[119] in 2018 [180] in 2019	M-ED algorithm MineAuth method		186,29Mb	46.85%	0.4%	500 samples	87,759 ms 20 - 60 min		
Power consumption Fusion	[86] in 2015 [134] in 2016 [121] in 2017	StrOUD k-NN BayesNet	532 ms 5.61 s						156	

and a hostile scenario and evaluated the performance of a gait authentication system based on wearable sensors. In both scenarios, they collected gait patterns by placing a wearable accelerometer sensor to the hip of test subjects. This method has been suggested, from the authors, for the unobtrusive authentication of users on mobile devices. During the friendly scenario, they experimented by launching a passive (zero-effort) attack. This means that individuals submitted their biometric characteristics as if they were trying to successfully verify themselves against their own templates, but in fact, their biometric characteristics were compared with a nonself template. Participants walked in their customary way and an EER of about 13% and a recognition rate of 73.2% was obtained by employing the averaged cycle method. During the second part they launched a minimal-effort impersonation attack where participants tried to imitate the walking manner of another individual after observation. Based on FAR error analysis the probabilities for impostors being successfully allowed access did not increase. Within the third part, they launched an attack based on the hypothesis that attackers were aware of their closest individual in the database. This means that attackers selected an individual whose gait was the most similar to theirs, which resulted in them becoming a great threat to the system.

Mjlaand et al. [133] extended the work of Gafurov et al. [142] by performing three differently structured scenarios, namely a friendly, a short-term and a long-term hostile that involved 50 individuals. The system's baseline performance within the friendly scenario was 6.2% in terms of EER. During the short-term hostile scenario, they selected seven individuals whose gait templates had a small distance between them. One of the participants acted like a victim while the other six tried to imitate him individually. The attackers' training lasted two weeks watching videos with the walking manners of the victim. The comparison between the gait templates of the victim and the attacker was made by employing the distance metric Dynamic Time Warping (DTW). Regarding the long-term hostile scenario, while the training was the same as in the short-term hostile scenario it was addressed to only one attacker and lasted six weeks. The conclusion on this part of the work was that the attacker's ability to learn how to walk like the victim does not improve even if he is long-term trained.

Kumar et al. [60] incorporated a digital treadmill in an attack scenario over a "baseline gait-based authentication system (GBAS)". First, they collected accelerometer readings to capture walking pattern from 18 individuals by using a smartphone. Then, they tested the performance of the GBAS on their dataset by using five different machine learning algorithms. The use of random forest achieved the best results, specifically the FAR was 6% and the FRR was 3%. Afterwards, they investigated the performance of the GBAS after three attack attempts. They trained impostors to imitate their victims by using a digital treadmill to control gait characteristics. Their attacks degraded GBAS' s performance significantly and achieved an increase in mean FAR regarding all attacks on all users for all classification algorithms. More specifically the increase in mean FARs were 544% for logistic regression, 549% for Bayes network, 580% for multilayer perceptrons, 590% for SVM and 742% for random forest. The random forest classifier had the most degraded performance where the FAR of 6% increased to 44.5%.

Muaaz and Mayrhofer [95] evaluated gait recognition security under a realistic imitation attack scenario. Five specialists in body movement mimicking acted as attackers and four smartphone users acted as victims. Attackers and victims were paired according to physical similarities such as age, weight, height, etc. The attack scenario was conducted in two different phases, the reenact and the coincide phase. During the reenact phase victims walked in normal pace with the smartphone placed inside their trousers pocket while attackers observed them. Following, attackers walked along with their paired victims to observe them more closely, and finally, they tried to mimic their victim's walking manner. During the coincide phase, the authors made an analysis on the attackers' imitation skills improvement while they walked close to their victims. In both phases the False Acceptance Rate

achieved was 0% showing that attackers could not achieve sufficient scores and be allowed access to the system. Moreover, when attackers imitated the walking style of their target person, they lost their steps pace and impersonation became much more difficult in 29% of the attempts.

Shrestha et al. [64] conducted a three-fold evaluation of the use of walking patterns for addressing the vulnerability of a Zero Interaction Authentication (ZIA) system. First, they designed and implemented a ZIA authentication system that used the sensors of a smartphone and a smartwatch to extract walking characteristics and authorize access. Then, they evaluated their authentication system under an imposter and a treadmill attack scenario. In the imposter attack scenario, the imposter possessed the legitimate user's smartphone and smartwatch and tried to mimic his walking pattern and fool the authentication system. In the treadmill attack scenario, the attacker used a treadmill to control gait characteristics and match the extracted features from the walking pattern of the legitimate user. Their results showed that when both devices were used, both attacks became hard to execute, achieving on average 4.55% FAR, even if the attacker's capabilities were high. Kumar et al. [134] in the authentication mechanism described previously evaluated the performance under the zero-effort or random attack without launching active attacks. The training of four classifiers included both genuine and impostor samples. In the intra-session environment, a 0% mean dynamic FAR and a 0% dynamic FRR was achieved by all mechanisms of authentication. In the inter-session environment, the best error rates achieved were 2.2% mean dynamic FAR and 4.2% dynamic FRR by the design based on feature level fusion with the k-NN classifier. In the inter-phase environment, the DFAR and DFRR were increased from 5.68% and 4.23% to 15.03% and 14.62% respectively by the deployment of the design based on fusion at feature level and k-NN classifier.

4.3. Attacks on keystroke dynamics

Several studies examine the vulnerability of authentication systems based on keystrokes. In this section we present four types of attacks including the Frog-Boiling attack [151], the Algorithmic attack [150], Mimic attacks [83,146], and the Snoop-forge-replay attack [131]. We have included studies that use virtual screen keyboards [146,150,151] as well as physical keyboards [83,131] because they are similar and exhibit very interesting findings.

The Frog-Boiling attack systematically alters the templates of target users by poisoning them via the template update mechanism to effectively direct them to a template which is selected by the attacker [47, 151]. The Frog-Boiling attack was used by Wang et al. [151] to exploit a template update mechanism and poison the keystroke dynamics system user templates. The attack increased the verifiers' EERs from between 9.9% and 18.9% to between 19.1% and 63.6% as it transformed well-performing users into ill-performing users. Attackers with weak destination templates caused higher EER increases compared to attackers with strong templates.

Serwadda and Phoha [150] investigated how a keystroke-based authentication system performed when exposed to synthetic attacks that were mimicking the legitimate user. They statistically analyzed keystroke data collected from more than 3000 users over 2 years to mimic the target user. Based on the observed statistical traits they designed and executed algorithmic attacks over three keystroke verification systems that used a password. The attack synthetically generated the keystroke sequence that corresponded to the target user's profile. Their algorithmic attack increased the Z-score, Scaled Manhattan and Naïve Bayes verifiers mean EERs by between 28.6% and 84.4% in comparison with the zero-effort attacks which are commonly used to evaluate the keystroke-based biometric systems performance.

Negi et al. [146] executed a scenario where an attacker gained access to the leaked authentication information of a website (username and password). Then the attacker tested if this authentication information

could be used on a banking website that incorporated a keystroke dynamics-based security layer. They experimented with two adversarial agents of their own design: Targeted K-means++, and Indiscriminate K-means++, on various datasets, and multiple state of the art classification algorithms. For Targeted K-means++ timing information about a given password was obtained simply by asking users on a paid crowdsourcing platform (Amazon Mechanical Turk) to type the specific password. Indiscriminate K-means++ was designed based on the idea that given the username and password timing vectors can be generated if general population timing data for each key is available. Timing data can be obtained from unsuspecting users by botnets and keyloggers. Their experiments demonstrated that their attack algorithms were effective across different settings. With Targeted K-means++ they succeeded within ten attempts to compromise the security in 40-70% of users. The Indiscriminate K-means++ compromised the security in 30-50% of users. Moreover, they showed that the K-means++ generalized well to touch screen swiping data.

Meng et al. [83] used an interface named mimesis which provided feedback to train impostors in imitating another individual's typing patterns through the adjustment of their typing patterns. They conducted their experiment with 84 individuals acting as impostors. They set a prerequisite that the impostors obtained the victim's typing patterns by two different ways, either by extracting them from a compromised database or by capturing keystroke samples as the victim authenticated using a keylogger. They used 2 eight-digit passwords of different difficulty, and the false acceptance rate (FAR) of the difficult and easy password increased from 20% and 24% respectively (before training), to 42% and 63% (after training with partial information of the victim). With full information of the victim, the FAR increased to 99% for both passwords for the 14 best attackers, showing that when a victim's typing pattern is known, imitation is possible.

Rahman et al. [131] presented an attack on continuous verification systems based on keystroke, named snoop-forge-replay. The attack's execution involves three steps: 1) stealing timing information of keystroke from a target user using a keylogger, 2) using the stolen keystroke timing information to forge a typing sample, and 3) replaying the forged typing sample in the continuous verification system. They launched the snoop-forge-replay attacks by replaying stolen samples into the authentication system. The average EERs achieved by the attack were between 0.487 and 0.912 depending on the number of stolen keystrokes. The attack increased EERs from between 69.33 to 2730.55% since the baseline EERs of the zero-effort attacks were between 0.03 and 0.285.

4.4. Attacks on touch dynamics

A considerable amount of research, for example [33,34,149], has argued that an authentication system based on touch could encounter two threats a form of shoulder surfing attack and the malware attack. Regarding shoulder surfing attacks Frank et al. [33] argued that an individual cannot learn the touch behavior of another individual simply by looking over his shoulder. Also, the aforementioned research has shown evidence that touch-based authentication systems hold a lot of promise for zero effort attacks [57] because the illegitimate user that gains access to the device is prevented by the mismatch between touch patterns. However, these attacks do not represent the entire field of threats the system could face for two reasons [59]:

- a Users' behavioral biometrics show a wide variance and overlapping between users [56,153]. In case we extract statistics from a database with large population records they could be used to generate forgeries and increase error rates [82,150].
- b The gestures on which continuous authentication is based can be easily implemented using commercially available robotic devices.

Serwadda et al. [59] demonstrated the efficient use of a "Lego" robot

Table 11
Practical attack techniques on behavioral biometrics.

Biometrics	Works	Attacks	Performance (%)		Compromise of security of users	EER	FRR
			FAR	Increase of system errors			
Walking gait	Gafurov [142]	Imitation				13	
	Mjalaand [133]	Imitation				6.2	
	Kumar [60]	Imitation	44.5				
	Muaaz [95]	Mimic	0				
	Shrestha [64]	Mimic	4.55				
Keystroke dynamics	Kumar [134]	Zero effort	DFAR 2.2				DFRR 4.2
	Wang [151]	Frog-Boiling				19.1, 63.6	
	Serwadda & Phoha [150]	Algorithmic		84.4% MEER			
	Negi [146]	Mimic			40-70, 30-50		
	Meng [83]	Mimic	99				
Touch dynamics	Rahman [131]	Snoop-forge-replay				48.5 - 91.2	
	Serwadda [59]	Robotic		1004% EER			
	Serwadda [144]	Robotic		70% FAR			

in creating forgeries that accomplish disturbingly high penetration rates over touch-based authentication systems when guided by input deriving from swiping statistics of the general population. They investigated the attack's impact by utilizing the best touch-based authentication classification algorithms and discovered that it augmented the classifiers' EERs by between 339% and 1004% depending on factors such as the failure-to-enroll threshold and the touch stroke type that the robot created. Their work raises doubt about the performance evaluating approach employed in touch-based authentication systems, namely the zero-effort impostor approach. In another study by Serwadda et al. [144] they used a robotic attack based on both population statistics and patterns of a specific user. For the attack driven by population statistics, they used patterns extracted from a large users' population, while for the attack towards a specific user they used stolen samples of a specific individual. By using seven verification algorithms they demonstrated that in both attacks the performance of a touch-based authentication system degrades significantly. For the algorithm that was the least affected, the population attack caused a greater than 70% increase in FAR.

In Table 11 we present the type of attacks and system performance after the attack. In some cases, we present the minimum and maximum of EER while in others the minimum and maximum of FAR and FRR. There are cases where we present the highest system error or the percentage of error increase after the attack. Finally, in some cases compromise of the security of users is also presented.

4.5. Discussion on practical attacks

Gait: As we previously saw the attacks on gait refer to mimic attacks including impersonation attacks of minimal effort and attacks after the adversary training. Research results on accelerometer-based authentication systems demonstrated that the impersonation attack of minimal effort on gait biometric will not increase the impostor odds of being accepted [142]. Moreover, even when attacks are launched after the long-term training of the adversaries, their ability to learn the target user's walking manner does not improve [133]. Imitation is a difficult task and even when professional actors who are specialists in body motions and body language mimicking and are of the same gender and other physical characteristics with the victim are trained as impostors, they fail in being accepted by the authentication system [95]. On the contrary, in the research of Gafurov et al. [143], it was reported that it is higher likely an impostor to be accepted when being of the same gender with the victim. In addition, attackers who knew who their nearest person in the database is became a threat to the authentication system [142]. Also, Kumar et al. [60] by launching a treadmill imitation attack increased the average FAR from 5.8% to 43.66% and characteristically report that: "The authentication systems based only on accelerometer

readings are vulnerable to imitation attacks". Of course, works like for example Shrestha et al. [64] and Kumar et al. [134], suggest a multi-modal gait biometrics solution using the accelerometer and the gyroscope. These two works use sensors which are placed at several parts of the body and authenticate users based on their walking patterns. They also showed that the fusion of these modalities improves the overall performance of the system and adds an additional layer of defence against mimicry attacks of high-effort. We believe that these two works provide serious evidence that they are a solution to the previous issue, but further research is necessary.

Keystroke: On keystroke-based authentication systems we saw the following types of attacks: The Frog-Boiling attack [151], the Algorithmic attack [150], Mimic attacks [83,146], and the Snoop-forge-replay attack [131]. Although, the performance of a biometric system is enhanced when the templates of users are often updated, little research has been conducted on attacks towards such update mechanisms. The Frog-Boiling attack is a synthetic attack that secretly exploits the update mechanism of users' templates to poison them. The attack not only is hard to detect but also has a great impact because it transforms users from well performing into ill performing and the error rates increase. In fact, it not only allows illegitimate access to the account of the victim, but the adversary also weakens the victim's template. This aspect makes the specific attack important, as the biometric system becomes vulnerable to intrusion from many impostors and for a long time [151].

The foundation for the employment of keystroke biometrics in authentication systems lays in the uniqueness property of typing patterns. A large body of research suggests that such patterns are unique and hard to imitate, thus the research effort in this area has focused on detection techniques to differentiate legitimate users from impostors. However, the imitation of the legitimate user's keystroke patterns can be done successfully by an impostor [83,146]. More specifically, impostors can be trained to imitate another individual through the adjustment of their typing pattern by using a training interface, given that they possess information on the victim's typing pattern. This information can be obtained either by a leaked database or by installing a keylogger [83].

In case an attacker wants to mimic a target user's typing pattern he can gather information in other ways, for example by collecting samples from other individuals rather than the target individual. Indeed, based on the hypothesis that the unique typing style of an individual belongs to a greater group of similar styles; the set of keystroke timing patterns can be clustered into a small number of clusters. Practically, this means that users with similar typing styles can be grouped to the same cluster. By gathering information from the general population using keyloggers or a crowdsourcing platform all such clusters can be created and mimic the typing patterns of the targeted individual [146]. The mimicking of a

target user can also be done by an algorithmic attack that statistically analyses the keystroke data to synthetically generate the keystroke sequence that corresponds to the target user's profile [150]. Finally, in the case of snoop-forge-replay attack only a small amount of the victim's stolen timing information using keyloggers was necessary to create forgeries and make the attack effective [131]. From the above stems the fact that attackers can easily obtain a lot of keystroke data and extract the statistics of keystroke feature in many different ways. Indeed, they can use bots, crowdsourcing platforms, unsuspecting users by fooling them or extracting statistics of feature directly from keystroke datasets that are publicly accessible. As we saw these attacks are highly effective and consequently, there is a need for research on technologies that can defend against the previously mentioned attacks.

Touch gestures: Several studies [33,34,149] have shown evidence that touch-based authentication systems are much promising regarding zero effort and shoulder surfing attacks. The installation of a malware application on the user's device could result in a more effective attack if the attacker has information on how to compute the features, since the malware can obtain and reveal the legitimate user's touch patterns to the attacker [33]. The case of a malware attack is quite likely to be addressed by using an anti-malware software. However, if the user does not have anti-malware, this attack is quite dangerous for the authentication system. Also, an adversary accessing population statistics could gain much information on the patterns of a particular user without needing a malware for monitoring him. In addition, sophisticated adversaries that can use advanced robots, may render the attack even more successful.

Due to the attacks, several privacy issues are emerging. Some of these issues relate to the behavioral biometrics themselves, for example, the leakage of biometrics from a malware, a statistical, a robotic, etc., attacks. Apart from that, additional privacy issues emerge when attacks are successful and users' personal data are exposed. For the above reasons, we believe that research on technologies that can defend against attacks that are based on population statistics as well as robotic attacks should continue.

5. Possible countermeasures on practical attacks on BB

There are three key ways for the enhancement of security in behavioral biometrics authentication:

Adding features: Various features from the user's behaviour are possible to be extracted and used for the enhancement of security in behavioural biometrics authentication systems. Feng et al. [149] extracted data of touch from smartphones equipped with a touchscreen from 40 users and obtained approximately 7.5% FAR at approximately 8% FRR using the random forest verifier. Afterwards, a digital glove was worn by users that provided additional information on their hands' movement. With the use of the digital glove they reduced FAR at 4.66% and FRR at 0.13% and enhanced security. Several studies showed that pressure features are extremely hard for an impostor to imitate. Zhao et al. [35] combined pressure features with six touch traces for mobile devices authentication and achieved an EER of 2.62%. Wolf et al. [128] combined data from the accelerometer, the touch screen and the keyboard and identified individuals at 83% by using a simple normal distribution method. In a study of Tasia et al. [67] where they combined several features, they concluded that the best results in user authentication were achieved when combining pressure with time features. Kumar et al. [145] experimented on fusing typing patterns, swiping gestures and the corresponding patterns of phone movement on a 28 users dataset. An accuracy of 93.33% was achieved by fusing patterns of swipe and the corresponding movement patterns of the phone at feature level. An 89.31 % accuracy was achieved by fusing behaviors of typing and the corresponding movement patterns of the phone at score-level.

Combination with other biometrics: Several combinations of behavioral biometrics with physiological biometrics are possible. Bigun et al. [68] evaluated the combination of fingerprint and voice data and achieved 0.94% EER. Morris et al. [69] applied the combination of voice, face and signature data and achieved an EER of 1%. Kim et al. [70] used voices and teeth images and had 2.13% EER. In another experiment, Kim et al. [71] combined face, teeth and voice and they achieved an EER of 1.64% confirming the improved performance of combining modalities in comparison to single modalities authentication methods.

Combination with non-biometrics: Approaches that combine behavioural biometrics with password or token-based authentication enhance the authentication accuracy and are thoroughly investigated in the literature. De Luca et al. [72] examined the shape of the users input, and more specifically unlocking the screen and password input, in combination with the way they performed the input. Their method achieved 77% accuracy resulting in an increased security. Sae-Bae et al. [73] used multi-touch gestures which included palm position, fingertip movement

Table 12
Possible countermeasures.

Method	Works	Combination	Performance (%)			
			FAR	FRR	Accuracy	EER
Adding features	Feng et al. [149]	Touch data and hand movement.	4.66	0.13		
	Zhao et al. [35]	Six touch traces with pressure features.				2.6
	Wolf et al. [128]	Accelerometer, touch screen, keyboard.			83	
	Tasia et al. [67]	Keystroke by adding the features of pressure and size.				
	Kumar et al. [145]	Swiping gestures, Typing patterns & Phone movement.			93.33 89.31	
Combination with other biometric characteristics	Bigun et al. [68]	Fingerprint and voice.				0.94
	Morris et al. [69]	Voice, face and signature.				1
	Kim et al. [70]	Teeth image and voices.				2.13
	Kim et al. [71]	Face, teeth and voice.				1.64
Combination with non-biometrics	De Luca et al. [72]	Screen pattern and password.			77	
	Sae-Bae et al. [73]	Palm position, fingertip movement and dynamic.			90	
	Shahzad et al. [74]	Finger velocity, device acceleration, and stroke time.				0.5
	Ohana et al. [75]	Built a key/lock system for the mobile phone and its charger.				
	Sun et al. [76]	Passwords inputs in combination with the way that the inputs are performed by the legitimate user.			97.5	

and fingertip dynamics. They achieved a 90% accuracy rate by using single gestures and discovered that the accuracy improved significantly when multiple gestures were performed in sequence. In a study of Shahzad et al. [74] they combined the velocity of the finger, the device acceleration, and the stroke time to examine how users perform the input, even when attackers saw what gesture a user performed, through shoulder surfing or smudge attacks, they could not reproduce his behavior. Their method achieved 0.5% average EER. Ohana et al. [75] built a key/lock system for the mobile phone and its charger, thus making the phone useless when separated from its power source. This system, in combination with biometrics would discourage the theft of the mobile phone. Sun et al. [76] used curves as passwords inputs in combination with the way that the inputs are performed by the legitimate user. Their system achieved an accuracy rate of 97.5%.

The above methods improve the performance of authentication systems but have not been tested against trained imposters. All the data presented in Table 12 relate to the known system performance indexes FAR, FRR etc., but they do not show the improvement of systems resistance to real attacks.

5.1. Discussion on countermeasures on practical attacks

As it is previously shown in Table 12 the performance of authentication can be improved by incorporating additional biometrics in the authentication system, i.e. the use of multimodal biometrics. Multimodal biometrics-based authentication produces more consistent and higher authentication accuracy in comparison with the single biometrics-based authentication and achieves upgraded performance [24]. Many research works and practical implementations widely adopted the multimodal-based framework since the reliable authentication of a high level cannot be guaranteed by the single biometric [5,9]. A plethora of studies have emphasized the superiority of multimodal biometric approaches to single biometric methods [77–81]. Since multimodal biometrics is performing better a well-built authentication mechanism is necessary to ensure the successful implementation of multimodal user authentication.

6. Lessons learned, challenges and future trends in this area

Following, we present lessons learned, challenges and future trends regarding BBCA technology.

6.1. Lessons learned

Behavioral biometrics and CA technology advantages: CA technology is a promising method with many advantages. It can perform without user intervention, it doesn't require additional hardware and it presents high accuracy for zero-effort evaluation approach.

Inconsistent behavior of the legitimate user: a major deficiency that results from the inconsistent interaction of users with their mobile phones is the unreliable performance [11]. In addition, a physical injury or a panic situation may result in behavioral changes, which can lead to inconsistent reactions by the legitimate user. Therefore, the appropriate management of unexpected situations must be established [52].

Changes in user behavior and habits: User behaviors and habits may change over time; thus, authentication systems should be able to adapt to these changes. The paradigm of Continual learning [163,167,168] can train successfully deep discriminative models as well as deep generative models in complex continual learning scenarios where new tasks appear, and existing tasks change over time. Therefore, this paradigm could be employed towards this direction.

Research data collection procedure: Many users avoid participating in time consuming, painstaking procedures for the collection of biometric features. This results in not fulfilling the data collection procedure.

Aspects of machine learning on mobile devices: Machine learning on mobile devices involves several aspects such as the amount of data from

each user that is necessary for each approach to be effective, the computational and communication cost, and the usage of battery, memory, and CPU. When an authentication approach is evaluated, these aspects should be weighted and reported. Most researchers do not measure the aforementioned aspects in their work; thus, we cannot make a comparison of each method based on these aspects. We believe that future research should measure those aspects to enable the comparison.

Evaluation metrics: research works should weight and include at least the following metrics: FAR, TAR, FRR, EER, and Accuracy when evaluating the performance of their approach. In this way, it will be feasible to make a comparison between the approaches.

Vulnerabilities resulting from population statistics and users clustering: an adversary can gather information from the general population and efficiently mimic the typing patterns of the targeted individual subsequently avoiding the process of monitoring him by using, for example, a malware [59,146,150]. The adversary can additionally employ advanced robots, to make the attack even more successful [59,144]. Also, the unique typing style of an individual belongs to a greater group of similar styles therefore the employment of adversarial agents such as the Targeted and the Indiscriminate K-means++ as shown in [146], can render the attack efficient.

Security in behavioural biometrics authentication: The primary menace is that the legitimate user's behaviour can be mimicked directly or indirectly by an impostor. Security in behavioural biometrics authentication can be improved if features are added or behavioural biometrics are combined with other biometrics or with non-biometrics.

6.2. Challenges

6.2.1. Behavioral Biometrics collection and feature extraction

Need for a user-friendly BB collection methodology: A major challenge is the design of a methodology for collecting behavioral biometrics, in a way that makes it user-friendly. The selection and optimization of a proper set of behavioral biometrics constitutes a challenge and an open problem.

Lack of real-world datasets: Another great challenge is the small number of real-world datasets. Besides, the availability of a public behavioral biometrics database for research purposes is necessary [10].

6.2.2. Systems evaluation

Maximizing accuracy: As we examined in the literature most studies implement approaches using machine learning algorithms but very few use deep machine learning. We consider that the accuracy of an authentication system based on behavioral biometrics must be tested with deep machine learning algorithms to examine if the performance can be increased.

Need for high-effort evaluation approach: Most of the works presented in this paper are using the zero-effort approach to evaluate the performance of their authentication systems. There is a need to direct the evaluation approach from the zero-effort to a high-effort to see the actual performance of the systems in the light of today's possible threats.

Changes in user behavior and habits: The paradigm of Continual learning [163,167,168] can be used to successfully train deep discriminative models as well as deep generative models in complex continual learning scenarios where new tasks appear, and existing tasks change over time. This will help towards the direction of solving the issue of authentication systems to adapting to changes in user behavior and habits.

6.2.3. Security and usability

Balance between security and usability: Behavioral biometrics authentication is subject to limitations, such as risk of false positives/false negatives, balance between security and usability, privacy concerns, etc., which result to its limited applicability. To overcome these limitations, it is crucial to maximize accuracy and examine how to find a

balance between security and usability [112,113].

Defense against Robotic attacks: Robotic attacks constitute an open problem on touch-based authentication systems [59,144]. Adversaries can execute an effective robotic attack with cheap, off the shelf equipment and basic knowledge of programming. When combined with access to population statistics the attack becomes even more effective. With this threat case in mind, we suggest that a thorough evaluation of touch-based authentication systems should commence and proceed with research on technologies that could defend against robotic attacks.

Defense against mimicking attacks: Keystroke-based authentication systems are vulnerable to mimicking attacks. Attackers can use bots, crowdsourcing platforms, unsuspecting users or publicly available datasets to access keystroke data, extract keystroke feature statistics and launch an attack. As these attacks are highly effective there is a need for research on technologies that can defend against such attacks.

Systems resistance to real attacks: In chapter 5 we have presented possible countermeasures on practical attacks techniques on behavioral biometrics. The methods presented indeed improve the performance of the authentication systems but do not show the improvement of the systems resistance to real attacks. We believe that the performance of systems against trained imposters should be tested.

Maximizing systems reliability: The performance of behavioral biometrics authentication can be enhanced with the combination of different modalities. The superiority of multimodal biometric approaches in comparison with single biometric methods is emphasized in a plethora of studies. These approaches must be further tested in real-world large datasets since they are considered as the most reliable and secure.

6.2.4. Technology acceptance

Determinants of end-users' acceptance: End users' perception and behavior constitute important considerations when designing systems since security, privacy and online identity management issues often trouble consumers. A study aimed at identifying non-technical issues, for example perceptions about the fears and expectations of future end-users, may be necessary for developing a strategy to support the acceptance of an innovation [147]. For these reasons, the assessment of technology acceptance determinants is crucial for addressing the problem of reduced use and utilization of the BBICA technology benefits.

6.3. Future trends

A large corpus of research has shown evidence regarding the superiority of the fusion of multimodal biometric approaches. A variety of combinations is used with the fusion of biometrics that achieves improved results compared to the single modality methods [24,53]. Thus, the fusion of multimodal biometrics authentication emerges as a definite future trend. There is also evidence that the fusion of multimodal biometrics in combination with the Zero Interaction Authentication (ZIA) paradigm may also be a trend.

6.4. Summary and conclusions

In the present survey, we investigate Behavioral Biometrics and Continuous Authentication technologies for mobile devices to offer an up-to-date, comprehensive, extensive and targeted survey. The review of a large corpus of research in the field of BBICA technology has shown that CA technology can provide high authentication accuracy. Though, most of these works have used the zero-effort evaluation approaches. We believe that there is a need to shift from the zero-effort evaluation to the high-effort evaluation to see the actual performance of the systems under the spectrum of today's possible threats. Behavioral biometrics are promising but also vulnerable to practical attack schemes such as malware, mimic, impersonation, spoofing, replay, statistical, algorithmic and robotics attacks. The incorporation of additional biometrics in the authentication system, i.e. the use of multimodal authentication,

is shown to improve security and guarantee reliable authentication. The superiority of multimodal biometric approaches to single biometric methods is emphasized in a plethora of studies. These approaches must be further tested in real-world, large datasets. Also, behavioral biometrics authentication is of limited extent due to some major shortcomings: risk of false positives/false negatives, the balance between security and usability, privacy concerns, etc. [1]. To overcome these shortcomings, it is critical to maximize accuracy and investigate how to balance security with usability. Since the behavior and habits of users may change over time, authentication systems should be able to adapt to these changes. In complex continual learning scenarios where new tasks appear, and existing tasks change over time the paradigm of Continual learning [163,167,168] could be employed to train successfully deep discriminative models as well as deep generative models. Therefore, this paradigm could constitute a solution. Also, most researchers do not measure the amount of data from each user that is necessary for each approach to be effective, the computational and communication cost, and the usage of battery, memory, and CPU in their work. Thus, we cannot make a comparison of each method based on these aspects. Future research should measure those aspects to enable the comparison. In addition, when researchers evaluate the performance of their approach, they should at least weight and report the following metrics: FAR, TAR, FRR, EER, and Accuracy. In this way, it will be feasible to make a comparison between the approaches. Finally, evaluating end-users' technology acceptance determinants is vital to addressing the problem of reduced use of BBICA technology. Finally, a major challenge is the testing of a methodology for extracting biometric features, in a way that makes it user friendly, because many users avoid participating in time consuming, painstaking procedures for the collection of biometric features. We hope that this article will trigger researchers and practitioners towards the investigation of open problems and challenges in this field.

CRedit authorship contribution statement

Ioannis Stylios: Conceptualization, Methodology, Investigation, Writing - original draft, Visualization, Funding acquisition. **Spyros Kokolakis:** Writing - review & editing, Supervision. **Olga Thanou:** Investigation, Writing - original draft, Visualization. **Sotirios Chatzis:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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